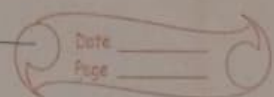
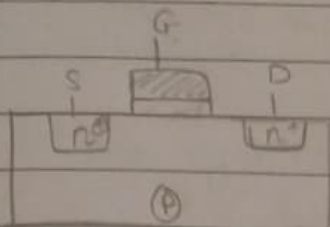


Analog signal $\rightarrow$ small letter.

Digital / DC signal $\rightarrow$ Capital $\rightarrow$



Working of n-channel MOSFET



$$V_{GS} = V_G - V_S$$

$$V_{DS} = V_D - V_S$$

$\rightarrow$ cutoff region.

i) $V_{GS} < V_{\text{thresh}}$ $\rightarrow$ No channel formation.

ii) $V_{GS} > V_{\text{threshold}}$ $\rightarrow$ n-type channel is formed.

iii) $V_{GS} > V_t$ $\rightarrow$ $V_{DS} < (V_{GS} - V_{th})$

$\rightarrow$ I_D flows from Drain to Source

$\rightarrow$ Ohmic region / Linear / Resistive.
Non-saturation region.

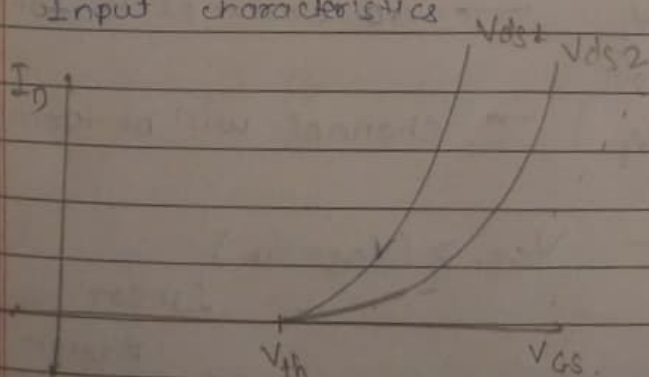
iv) $V_{GS} > V_t$ $V_{DS} \geq (V_{GS} - V_{th})$

$\rightarrow$ Drain current saturates (I_D constant)

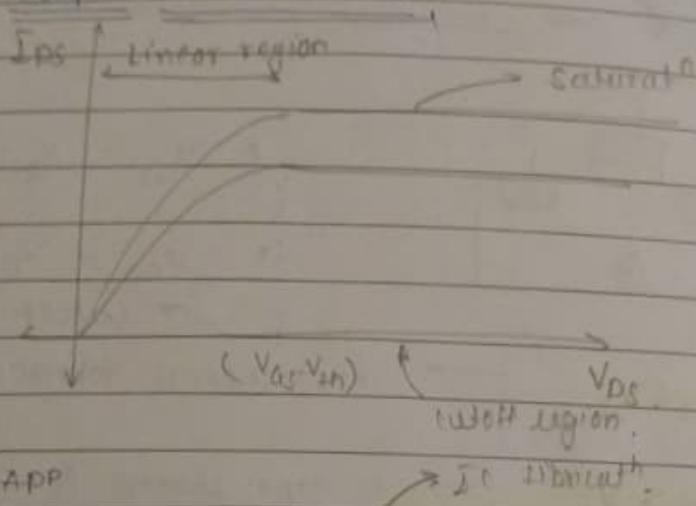
$\rightarrow$ channel gets pinched off.

$\rightarrow$ Saturation region.

Input characteristics



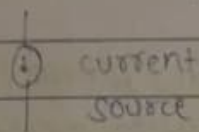
Output characteristics :-



APP

- 1) Acts as resistor in linear region
- 2) Constant current source in saturation region.

Avoid using inductors & resistors while designing chipset.

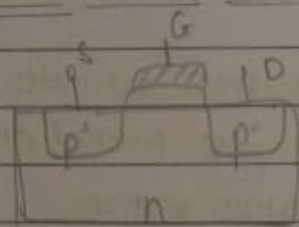


current
source



voltage
source.

P-channel MOSFET :-



- i) $V_{gs} > V_t$ $\rightarrow$ no channel is formed.
- ii) $V_{gs} < V_t$ $\rightarrow$ channel will be formed.
- iii) $V_{gs} > (V_{gs} - V_{th})$ Linear Region.

iv) $V_{GS} < V_{th}$

$V_{DS} \leq (V_{GS} - V_{th})$

Saturation Region

Q. Find the region of operation of a p-channel MOSFET

$V_t = -1V$ when it is biased such as

$V_{GS} = -5V$ $V_S = -1V$ $V_D = -6V$

~~$V_{GS} = V_G - V_S$~~

$V_{GS} = V_G - V_S$

$= -5 - (-1) = -4V$

~~$V_t = -1V$~~

$V_t = -1$

$V_{GS} < V_t$

Transistor is ON

$V_{GS} - V_t = -4 - (-1) = -3$

$V_{GS} \leq V_{GS} - V_{th}$

Transistor is in saturation region

Q. If $V_L = 0.4V$, $V_G = 0.4V$, $V_S = 1V$

$V_D = 0.9V$ for n channel

$V_{GS} = 0.4 - 1 = -0.6$

$V_t = 0.4V$

$V_{GS} < V_t$

Q. $V_T = 0.4 \text{ V}$, $V_G = 1.5 \text{ V}$, $V_S = 0 \text{ V}$
 $V_D = 0.5$ n channel

$$V_{GS} = 1.5 - 0 = 1.5 \text{ V}$$

$$V_T = 0.4$$

$$V_{GS} > V_T$$

$$V_{GS} - V_{th} = 1.5 - 0.4 = 1.1 \text{ V}$$

$$V_{DS} = V_D - V_S = 0.5 \text{ V}$$

$$V_{GS} - V_{th} > V_{DS}$$

∴ Ohmic Region

Length of channel = size of transistor

Second order effect :- comes to picture when we try to reduce channel length.

Drain Current Derivation :-

$$I_D = \frac{\mu_n}{2} (V_{GS} - V_{th})^2 \rightarrow \text{saturation region}$$

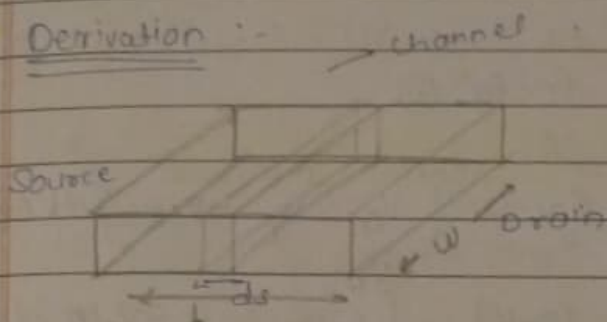
$$I_D = \mu_n \left[(V_{GS} - V_{th}) \cdot V_{DS} - \frac{V_{DS}^2}{2} \right] \rightarrow \text{Linear Region}$$

$$\mu_n = \frac{q}{m} \left(\frac{W}{L} \right) \rightarrow \text{aspect ratio}$$

mobility oxide capacitance

~~Oxide~~ Capacitance is Gate cap./unit area.

Derivation :-



$$C_{ox} = \frac{3.9 \epsilon_0}{t_{ox}}$$

oxide thickness

consider a small strip in channel,

$$\text{Capacitance offered} = C_{ox} \cdot w \cdot dx$$

$$I = \frac{dQ}{dt} \quad Q = CV$$

$$I = \frac{d(CV)}{dt}$$

$$I = C_{ox} \cdot w \cdot \frac{dx}{dt} \left[\right]$$

where $V(x)$ is the voltage applied at the strip.

$$I = \frac{dQ}{dt} \quad Q = CV$$

$$\therefore dQ = C_{ox} \cdot w \cdot dx [V_{GS} - V_t - V(x)] \quad \text{--- (1)}$$

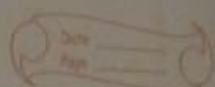
$$\rightarrow \frac{dQ}{dt} = \frac{dQ}{dx} \cdot \frac{dx}{dt}$$

$$= C_{ox} \cdot w \cdot [V_{GS} - V_t - V(x)] \cdot \frac{dx}{dt}$$

$$\frac{dx}{dt} = \mu_n \frac{dV(x)}{dx}$$

↓
mobility.

Value of C_{ox} in Quiz .



$$\frac{dQ}{dt} = C_{ox} \cdot W \cdot [V_{GS} - V_t - V(x)] \mu_n \frac{dV(x)}{dx} \quad (2)$$

Sub. Eqn (1) in Eqn (2) .

$$\therefore I = \mu_n C_{ox} W [V_{GS} - V_t - V(x)] dV(x)$$

$$\int_0^L I \cdot dx = \mu_n C_{ox} W \int_0^{V_{DS}} (V_{GS} - V_t - V(x)) dV(x)$$

$$I \cdot L = \mu_n C_{ox} W \left[\int_0^{V_{DS}} (V_{GS} - V_t) dV(x) - \int_0^{V_{DS}} V(x) dV(x) \right]$$

$$I = \frac{\mu_n C_{ox} W}{L} \left[(V_{GS} - V_t) V_{DS} - \frac{V_{DS}^2}{2} \right] \quad (3)$$

* Amount of charge stored in channel Q_c

for Saturation Region, we know,

$$V_{DS} = V_{GS} - V_t$$

$\therefore$ Substituting it into Eqn (3),

$$I = \frac{\mu_n C_{ox} (W)}{L} \left[\frac{V_{GS} - V_t}{2} \right]^2$$

$$\mu_n > \mu_p$$

$$K' = \mu_n C_{ox}$$

$$K_n = \mu_n C_{ox} \left(\frac{W}{L} \right)$$

$$V_{\text{overdrive voltage}} = V_{GS} - V_t$$

→ channel length

Q. for 0.8 process, $t_{ox} = 20 \text{ nm}$, $\mu_n = 100 \text{ cm}^2/\text{V}\cdot\text{sec}$
micro meter.

find C_{ox} , K' , V_{ov} req. to operate a transistor, with aspect ratio of 20 in saturation region Drain current = 1 mA . find, min. value of V_{DS} .

$$C_{ox} = \frac{\epsilon_0 \cdot 3.9}{0.8 \times 20} = \text{cancel out}$$

$$= \frac{8.85 \times 10^{-12}}{20 \times 10^{-9}}$$

$$= 1.72575 \times 10^{-3} \text{ F}/\mu\text{m}^2$$

$$K' = \mu_n C_{ox} = 100 \times 1.72575 \times 10^{-3} \text{ C}$$

$$= 0.173 \text{ A/V}^2$$

$$= 173 \text{ mA/V}^2 //$$

$$V_{ov} = \sqrt{V_{ov}} =$$

$$I_D = \frac{K_n}{2} (V_{ov})^2$$

$$1 \text{ mA} = \frac{K_n}{2} (V_{ov})^2$$

$$(V_{ov})^2 = \frac{1 \times 10^{-3} \times 2}{K_n}$$

$$① C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9 \epsilon_0}{t_{ox}} = 1.72 \text{ pF}/\mu\text{m}^2$$

$$② K'_n = \mu_n C_{ox} = 1.725 \times 10^{-5} \text{ mA/V}^2$$

$$③ I_D = \frac{K'_n}{2} \left(\frac{W}{L}\right) \times (V_{ov})^2$$

$$\therefore V_{ov} = 2.406 \rightarrow V_{DS}(\text{min.})$$

Transconductance %

$$g_m = \frac{I_D}{V_{GS}} \quad \text{--- ①}$$

Transconductance is ratio of output current to input voltage.

$$** \quad g_m = \frac{\partial I_D}{\partial V_{GS}} \quad \text{--- ②}$$

Transconductance is defined in saturation region.

$$I_D = \frac{K'_n}{2} \left(\frac{W}{L}\right) (V_{ov})^2$$

$$\frac{\partial I_D}{\partial V_{GS}} = \frac{K'_n}{2} \left(\frac{W}{L}\right) 2 \cdot (V_{GS} - V_t)$$

$$\therefore \boxed{g_m = K'_n \left(\frac{W}{L}\right) (V_{GS} - V_t)}$$

→ Type 1

Type ② :-

$$K_n' \left(\frac{W}{L} \right) = \frac{2 I_D}{(V_{GS} - V_{th})^2}$$

Substituting it into eqn ①,

$$g_m = \frac{2 I_D}{(V_{GS} - V_{th})}$$

$$g_m = \frac{2 I_D}{V_{OV}}$$

Type ③

$$V_{GS} - V_{th} = \sqrt{\frac{2 I_D}{K_n' \left(\frac{W}{L} \right)}}$$

$$\therefore g_m = K_n' \left(\frac{W}{L} \right) \left[\sqrt{\frac{2 I_D}{K_n' \left(\frac{W}{L} \right)}} \right]$$

Summary :-

→ transistor parameter.

$$① \quad g_m = K_n' \left(\frac{W}{L} \right) \cdot V_{OV}$$

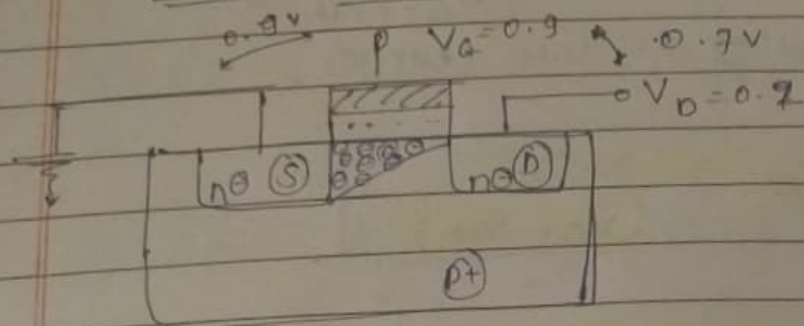
→ aspect ratio.

$$② \quad g_m = \frac{2 I_D}{V_{OV}} \quad \rightarrow \quad \text{Never use this formula unless no data is given}$$

$$③ \quad g_m = \sqrt{2 K_n' \left(\frac{W}{L} \right) I_D}$$

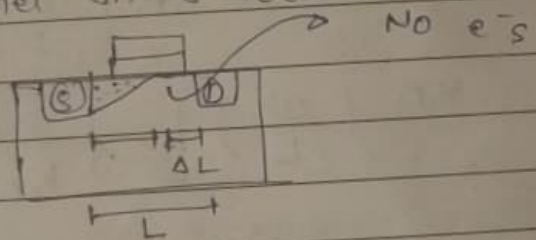
Second Order Effect:-

① Channel Length Modulation:-



since $V_{DS} < V_{GS} \rightarrow$ channel length will start decreasing at Drain.

As we start increasing Drain voltage channel length decreases and pinch off ~~decreases~~ increases, channel shifts towards source.



$$\begin{aligned}
 I_D &= K_n' \left(\frac{W}{L} \right) \times (V_{GS} - V_{th})^2 \\
 &= K_n' \frac{W}{L - \Delta L} (V_{GS} - V_{th})^2 \\
 &= K_n' \frac{W}{L} \frac{1}{\left(1 - \frac{\Delta L}{L}\right)} (V_{GS} - V_{th})^2
 \end{aligned}$$

$\frac{\Delta L}{L}$ is very small,

$$\therefore \frac{1}{1 - \frac{\Delta L}{L}} \approx 1 + \frac{\Delta L}{L}$$

$$\therefore I_D = K_n' \frac{W}{L} \left(1 + \frac{\Delta L}{L}\right) (V_{GS} - V_{th})^2$$

Derivation
not for
exam.

Date _____
Page _____

We know that ΔL is dependent upon V_{DS} as we increase V_D ΔL also increases

$$\therefore \Delta L \propto V_{DS}$$

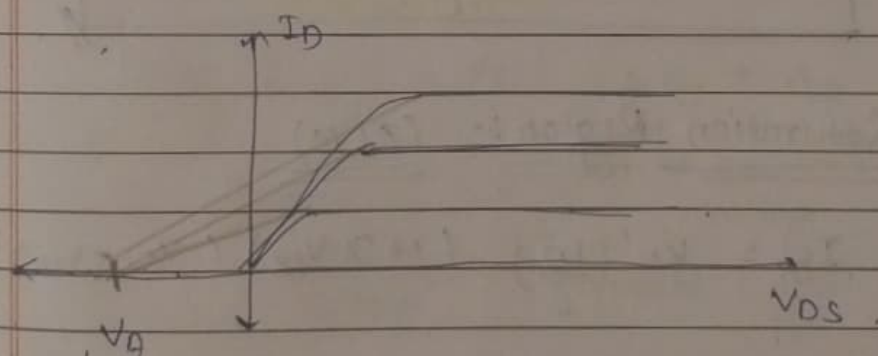
$$\therefore \Delta L = \lambda' V_{DS} \quad \lambda' \rightarrow \text{process constant}$$

$$I_D = \frac{K_n'}{2} \left(\frac{W}{L} \right) \left(1 + \frac{\lambda'}{L} V_{DS} \right) (V_{GS} - V_{th})^2$$

$$\text{Also } \frac{\lambda'}{L} = \lambda$$

$$\therefore I_D = \frac{K_n'}{2} \left(\frac{W}{L} \right) (1 + \lambda V_{DS}) (V_{GS} - V_{th})^2$$

If $\lambda = 0$, then CLM will not come in picture.



Early voltage $\rightarrow$ -ve V_{DS} given for zero I_D .
found by extrapolating I_D by applying -ve V_{DS} .

$$I_D = 0 = \frac{K_n'}{2} \left(\frac{W}{L} \right) (1 + \lambda V_{DS}) (V_{GS} - V_{th})^2$$

$$\therefore 1 + \lambda V_{DS} = 0$$

$$\lambda V_{DS} = -1$$

$$V_A = \frac{1}{\lambda}$$

$$-V_{DS} = \pm V_A$$

Resistance offered by Transistor

1) Ohmic Region (Linear Region)

$\rightarrow V_{DS}$ is small $\therefore$ we ignore

$$I_D = K_n' \left(\frac{W}{L} \right) (V_{GS} - V_{th}) V_{DS} \quad \frac{V_{DS}^2}{2}$$

$$r_{DS} = \frac{\partial V_{DS}}{\partial I_D}$$

$$\therefore \frac{\partial I_D}{\partial V_{DS}} = K_n' \left(\frac{W}{L} \right) (V_{GS} - V_{th})$$

$$\therefore r_{DS} = \frac{1}{K_n' \left(\frac{W}{L} \right) (V_{GS} - V_{th})}$$

2) Saturation Region:- (CLM)

$$I_D = K_n' \left(\frac{W}{L} \right) (1 + \lambda V_{DS}) (V_{GS} - V_{th})^2$$

$$\therefore \frac{\partial I_D}{\partial V_{DS}} = \underbrace{K_n' \left(\frac{W}{L} \right) (V_{GS} - V_{th})^2}_{I_D} \cdot \lambda$$

$$\therefore r_{DS} = \frac{1}{\lambda I_D}$$

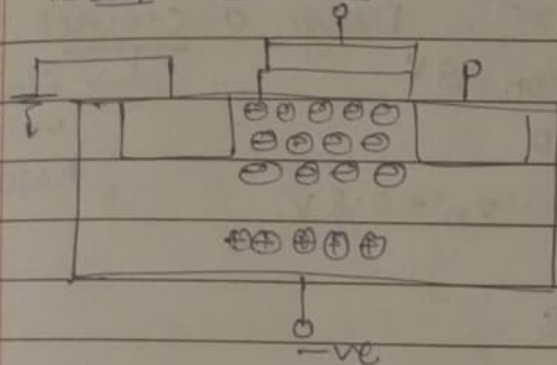
$$\therefore r_{DS} = \frac{V_A}{I_D}$$

Note:- If $\lambda = 0$, $r_o \rightarrow \infty$.

i.e., when there is no CLM do not consider r_o .

Second Order Effect:-

② Body Effect:-



Variation of threshold voltage with variation of substrate voltage.

As voltage of substrate decrease it attracts the charge and pushes $\ominus$ charge and thus channel length widens. $\therefore V_{th}$ increases & we see body effect.

$$V_t = V_{to} + \gamma \left[\sqrt{2\phi_f + V_{sb}} - \sqrt{2\phi_f} \right]$$

$\downarrow$ Threshold voltage $\downarrow$ B.E. parameter $\downarrow$ Fermi voltage $\downarrow$ $V_s - V_{Bulk}$ substrate voltage

Transistor Application :-

① Transistor as Switch

② Transistor as Amplifier

No A/C input is given

enters cutoff region & linear region

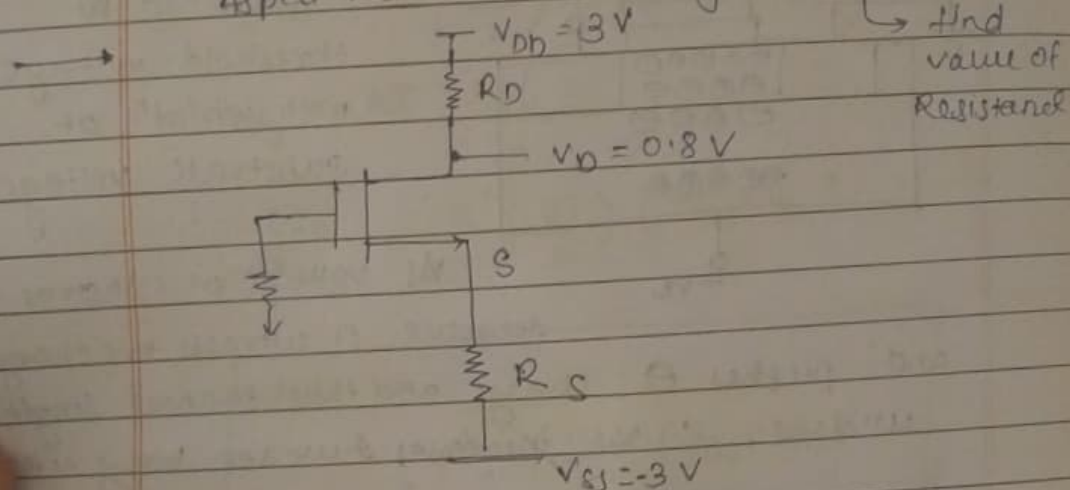
border blue linear & saturation

not exactly linear

otherwise it will be resistor

① Transistor as Switch :-

Q. $I_D = 1 \text{ mA}$, $V_t = 0.6 \text{ V}$, $\mu_n C_{ox} = 100 \frac{\mu\text{A}}{\text{V}^2}$
 $V_{DD} = 3 \text{ V}$, $V_{GS} = +3 \text{ V}$, $V_D = 0.8 \text{ V}$,
 Aspect ratio = 20. Design a circuit.



$$\frac{W}{L} = 20$$

$$\frac{L}{W} = \frac{1}{20}$$

$$R_D = \frac{V_{DD} - V_D}{I_D}$$

$$= \frac{3 - 0.8}{1 \text{ mA}}$$

$$= 2.2 \text{ K}\Omega$$

$$I_D = K_n' \left(\frac{W}{L} \right) (V_G - V_S - V_t)^2$$

$$I_D = K_n' \left(\frac{W}{L} \right) (-V_S - V_{th})^2$$

$$(-1)^2 (V_S + V_{th})^2 = \frac{I_D \times L}{K_n' W}$$

$$V_S + V_{th} = \sqrt{\frac{I_D L}{K_n' W}}$$

$$V_S = \sqrt{\frac{I_D L}{K_n' W}} - V_{th}$$

$$V_S =$$

$$I_{DQ} = \frac{100 \mu A}{2} \times 20 (-V_S - 0.6)^2$$

$$\frac{1}{1000} = (-V_S - 0.6)^2$$

$$-V_S - 0.6 = \sqrt{\frac{1}{1000}} = 0.0316$$

$$I_{DQ} = \frac{100 \mu A}{2} \times 20 (-V_S - 0.6)^2$$

$$(-V_S - 0.6)^2 = 1$$

$$V_S = -1.6$$

$$\therefore \frac{V_S - V_{SS}}{I_D} = R_S$$

$$\frac{-3 + 1.6}{I_D} = R_D$$

$$\therefore R_D =$$

Q Design a circuit and find the max. R_D to operate the device in saturation. Assume $V_{DD} = 6V$, $V_p = 3V$, $K_p' \left(\frac{W}{L}\right) = 1 \text{ mA/V}^2$, $V_{\text{Drain}} = 3V$ & $|V_t| = 1V$,

$$I_D = 0.5 \text{ mA}$$

$$I_D = \frac{K_p' \left(\frac{W}{L}\right) (V_{ov})^2}{2}$$

Note:- for a pmos transistor V_{ov} & V_t will be -ve.

$$\therefore 0.5 = \frac{1}{2} (V_{ov})^2$$

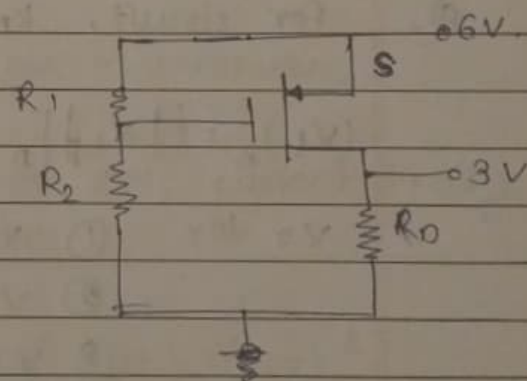
$$(V_{ov})^2 = 1$$

$$V_{ov} = -1$$

$$V_{ov} = V_G - V_S - V_t$$

$$-1 = V_G - 6 - (-1)$$

$$\therefore V_G = 4V$$



Using voltage divider rule,

$$V_G = \frac{6 R_2}{R_1 + R_2}$$

$$\frac{4}{6} = \frac{R_2}{R_1 + R_2}$$

$$R_2 = 4 \text{ M}\Omega \quad \text{Assume}$$

$$R_1 = 2 \text{ M}\Omega$$

For pmos to be in saturation condition,

$$V_{DS} < V_{GS} - V_t$$

$$V_D < V_G - V_t$$

$$I_D R_D < 4 + 1$$

$$I_D R_D < 5$$

$$\therefore R_D < \frac{5}{0.5 \text{ mA}}$$

$$\therefore R_D < 10 \text{ K}\Omega \quad \text{--- (1)}$$

From circuit,

$$I_D R_D = 3$$

$$\therefore R_D = 6 \text{ K}\Omega$$

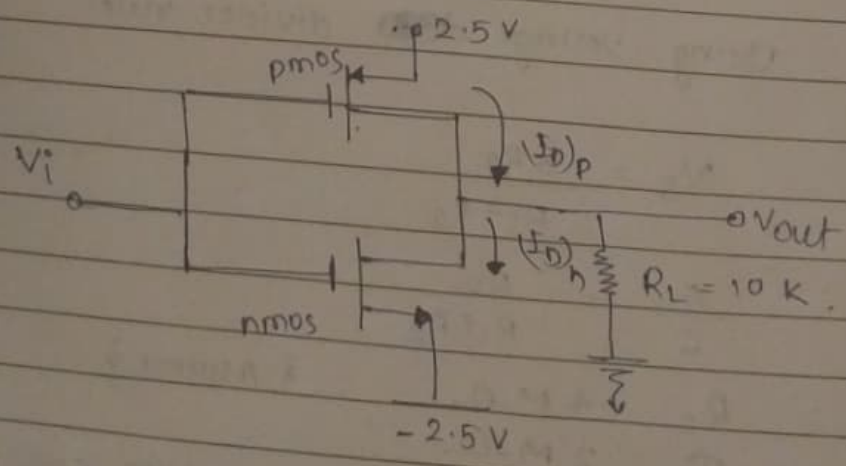
R_D of 6 K to 10 K will retain the transistor in saturation region.

Q. For circuit, $k_n' \left(\frac{W}{L}\right)_n = k_p' \left(\frac{W}{L}\right)_p = 1 \text{ mA/V}$
 $(V_{t_n})_n = (|V_{t_p}|)_p = 1 \text{ V}$. find $(I_D)_p$, $(I_D)_n$,
 V_O for

(1) $V_i = 0 \text{ V}$

(2) $V_i = 2.5 \text{ V}$

(3) $V_i = -2.5 \text{ V}$



(i) $V_i = 0V$

Nmos:-

for working of transistor,

$$V_{GS} > V_t$$

$$V_G - V_S = 0 - (-2.5) = 2.5$$

$$V_t = 1V$$

$$\therefore V_{GS} > V_t \quad \therefore \text{Nmos is ON.}$$

Pmos:-

for working of transistor

$$V_{GS} < V_t$$

$$\rightarrow -2.5 < -1 \quad \therefore \text{pmos is ON.}$$

$$\rightarrow V_{out} = 0$$

Both the transistors are in saturation.

$\therefore$ Both the transistors are ON, current directly flows from pmos to nmos.

$$\therefore (I_D)_p = (I_D)_n = \frac{K_n'}{2} \left(\frac{W}{L} \right) \cdot (V_{ov})^2$$

$$= \frac{1}{2} (2.53 - 1)^2$$

$$= \frac{1}{2} \times$$

$$= 1.125 \text{ mA}$$

ii) $V_i = +2.5$

$$V_G = 2.5V$$

for nmos:-

$$V_{GS} = 2.5 + (2.5) = 5V$$

$$\therefore V_{GS} > V_t \quad \therefore \text{nmos is ON.}$$

for pmos,

$$V_{GS} = V_G - V_S = 2.5 - 2.5 = 0,$$

$$V_t = -1$$

$$V_{GS} > V_t \quad \therefore \text{pmos is off!}$$

nmos:-

B Drain voltage varies from -2.5 to $0V$

$$\therefore V_{DS} < V_{GS} - V_{th}$$

$$V_{DS} < 5 - 1 < 4$$

$\therefore$ nmos is in Linear Region.

$$\therefore (I_D)_n = K_n' \left(\frac{W}{L} \right) [(V_{GS} - V_{th})(V_D - V_S)]$$

B from circuit,

$$V_{DS} = I_D \times 10K (-1) \quad \text{current is flowing from ground}$$

$$\therefore V_D = -10I_D \quad \text{to -ve.}$$

~~To find I_D and V_{out}~~

$$I_D = 1 \left(4 (-10I_D - (-2.5)) \right) \\ = -40I_D + 10.$$

$$I_D = 0.2439 \text{ mA.}$$

$$\frac{0 - V_{out}}{R_L} = I_D$$

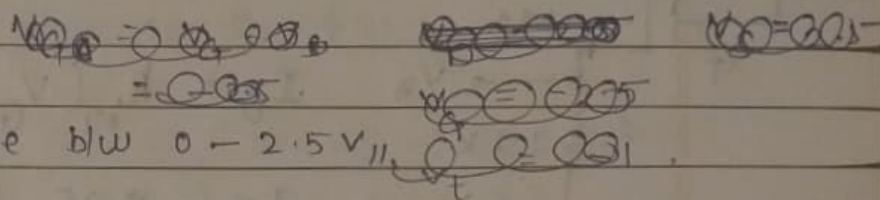
$$\therefore V_{out} = -2.44V.$$

iii) $V_1 = -2.5V$

Pmos $\rightarrow$ ON

nmos $\rightarrow$ OFF

pmos : $V_{GS} - V_t = -4V$



V_{DS} will be b/w $0 - 2.5V$

$\therefore V_D > V_G - V_t$
transistor is in Linear Region

$$I_D = K_p' \left(\frac{W}{L} \right) \left[(V_{GS} - V_{th}) \cdot V_{DS} \right]$$

$$I_D = 1 \left[-5 + 1 (I_D \times 10 - 2.5) \right]$$

$$\therefore I_D = -4 (10 I_D - 2.5)$$

$$\therefore I_D = -40 I_D + 10$$

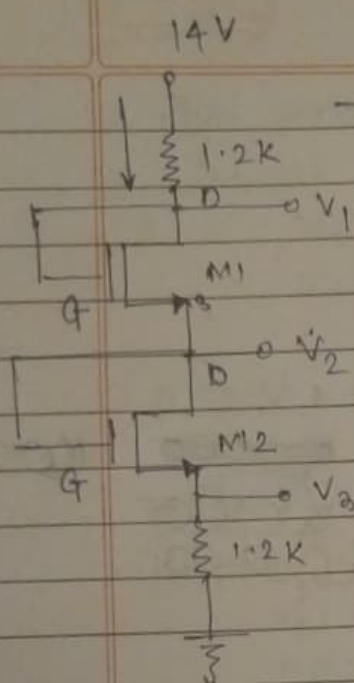
$$\therefore I_D = 0.2489$$

$$V_{out} = \frac{V_{out} - 0}{R_L} = I_D$$

$$\therefore V_{out} = I_D R_L = 2.44V$$

Q. Find V_1, V_2 & V_3 if two transistors are identical

$$V_t = 1V \quad K_n' \left(\frac{W}{L} \right) = 0.25 \text{ mA/V}^2$$



Diode transistor config.
→ Gate & Drain is shorted
→ Always in saturation.

$$14 - V_1 = I_D R_D$$

$$14 - V_1 = I_D (1.2) \quad \text{--- (1)}$$

$$I_D = k_{M1} (V_{G1} - V_{th})^2 \quad \text{for } M1$$

$$I_D = \frac{0.25}{2} (V_1 - V_2 - 1)^2 \quad \text{--- (2)}$$

for M2 :-

$$I_D = \frac{0.25}{2} (V_2 - V_3 - 1)^2 \quad \text{--- (3)}$$

$$V_3 - 0 = I_D (1.2) \quad \text{--- (4)}$$

$$V_1 - V_2 - 1 = V_2 - V_3 - 1$$

$$\frac{V_1 + V_3}{2} = V_2 \quad \text{--- (5)}$$

$$V_3 = 1.2 I_D \quad \text{--- (6)}$$

$$14 - 1.2 I_D = V_1 \quad \text{--- (2)}$$

$$I_D = \frac{0.25}{2} (14 - 1.2 I_D - 1.2 I_D - 1)^2$$

$$14 - \frac{1.2 I_D}{2} + \frac{1.2 I_D}{2} = V_2$$

$$V_2 = V_2$$

$$I_D = \frac{0.25}{2} (14 - 1.2 I_D - 1.2 I_D - 1)^2$$

$$2I_D = 0.25 (14 - 8 - 1.2 I_D)$$

$$2I_D = 0.25 (6 - 1.2 I_D)$$

$$= 0.25 \times 6 - 0.25 \times 1.2 I_D$$

$$(2 + 0.25 \times 1.2) I_D = 0.25 \times 6$$

$$2.3 I_D = 1.5$$

$$I_D = 0.6521$$

$$2I_D = 0.25 (14 - 8 - 1.22 I_D)^2$$

$$2I_D = 0.25 (6 - 1.2 I_D)^2$$

$$2I_D = 0.25 (36 + 1.44 I_D^2 - 2 \times 1.2 I_D \times 6)$$

$$2I_D = 0.25 (36 + 1.44 I_D^2 - 14.4 I_D)$$

$$2I_D = 9 + 0.36 I_D^2 - 3.6 I_D$$

$$0.36 I_D^2 - 5.6 I_D + 9 = 0$$

$$I_D = 1.82 \text{ A}$$

$$= 13.73$$

Take lower value always

$$V_1 = 14 - 1.2 I_D$$

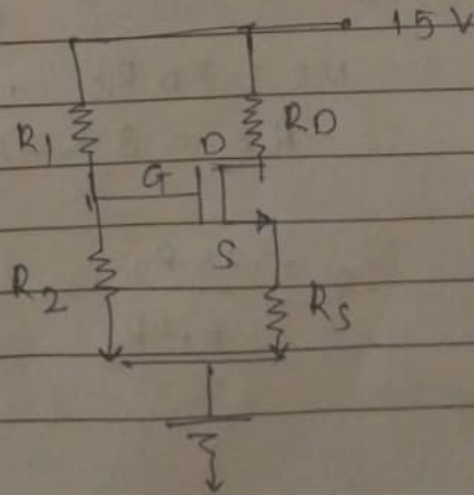
$$= 11.81$$

$$V_3 = 2.18 \text{ V}$$

Q. Design a circuit to obtain $I_D = 0.5 \text{ mA}$, $V_S = 4 \text{ V}$, $V_{DD} = 15 \text{ V}$, $V_{DS} = 5 \text{ V}$, $V_t = 1 \text{ V}$, $K_n' \left(\frac{W}{L} \right) = 1 \text{ mA/V}^2$.

Calculate the % change in I_{DQ} obtained when the MOSFET is replaced with another MOSFET having the same $K_n' \left(\frac{W}{L} \right)$ but with

$$V_t = 1.5 \text{ V}$$



Assume

$$V_{DS} = 5V$$

$$V_D = V_S = 5V$$

$$V_D = 9V$$

$$V_D = 9V$$

$$I_D = \frac{K_n'}{2} \left(\frac{W}{L} \right) (V_{OV})^2$$

$$\therefore 0.5 = \frac{1}{2} (V_{OV})^2$$

$$\Rightarrow V_{OV} = 1$$

$$V_{GS} - V_t = 0.7071$$

$$V_G - V_S - V_t = 0.7071$$

$$V_G = 0.7071 + 4 + 1$$

$$V_{OV} = 1V$$

$$V_G - 4 - 1 = 1V$$

$$V_G = 6V$$

$$V_G = 6V$$

$$15 - 9 = I_D R_D$$

$$\therefore 6 = 0.5 R_D$$

$$R_D = 12K\Omega$$

$$V_C = I_D R_C$$

$$R_C = 8K\Omega$$

$$V_G = \frac{15 R_2}{R_1 + R_2}$$

If MOSFET is pmos then take -ve value

$$\frac{6}{15} = \frac{R_2}{R_1 + R_2}$$

Let's assume $R_2 = 6 \text{ M}\Omega$

$$R_1 = 9 \text{ M}\Omega \parallel 1$$

New MOSFET

$$I_{D'} = \frac{K_n' (W/L)}{2} (V_G - V_S - V_L)^2$$

$$I_{D'} = \frac{K_n' (W/L)}{2} (6 - 8I_D - 1.5)^2$$

($\because V_S$ is current dependent)

$$I_{D'} = \frac{1}{2} (4.5 - 8I_D)^2$$

$$2I_{D'} = 20.25 + 64I_D^2 - 72I_D$$

$$64I_D^2 - 74I_D + 20.25 = 0$$

$$I_D = 0.7116$$

$$I_D = 0.4446 \quad \rightarrow \text{Take lower value}$$

$$\frac{\Delta I_D}{I_D} \quad \% \text{ change} = \frac{\Delta I_D}{I_D} = \frac{0.5 - 0.4446}{0.5} \times 100 = 11.08\%$$

Q. A MOSFET, whose $\lambda = 0.02/\text{V}$ carried a $I_D = 12 \text{ mA}$ when $V_{DS} = 5 \text{ V}$. If V_{DS} is increased to 10 V $I_D = ?$

$$\frac{I_{D2}}{I_{D1}} =$$

$$I_{D1} = \frac{K_n' (W/L)}{2} V_{ov} (1 + \lambda V_{DS1}) \quad \text{--- (1)}$$

$$I_{D2} = \frac{K_n' (W/L)}{2} (V_{OV})^2 (1 + \lambda V_{DS2})$$

Dividing ② by ①

$$\frac{I_{D2}}{I_{D1}} = \frac{1 + \lambda V_{DS2}}{1 + \lambda V_{DS1}}$$

$$I_{D2} = \frac{1.2 (1 + 0.02 \times 5)}{1 + 0.02 \times 10} = 1.1 \text{ mA}$$

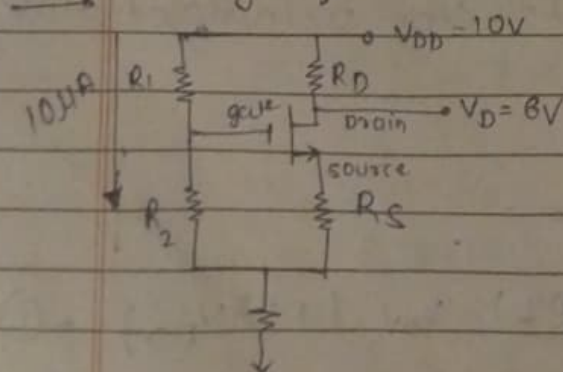
Q. For a MOSFET if W is increased 16 times while keeping I_D constant. Its g_m will increase by

$$g_{m1} = \sqrt{2 I_D K_n' \left(\frac{W}{L}\right)}$$

$$g_{m2} = \sqrt{2 I_D \times 16 K_n' \left(\frac{W}{L}\right)}$$

$$\therefore g_{m2} = 4 g_{m1}$$

Q. Design a circuit so that it operates in saturation, with $I_D = 2 \text{ mA}$, $V_d = 6 \text{ V}$, $V_{DD} = 10 \text{ V}$, $K_n' (W/L) = 0.25 \text{ mA/V}^2$, $V_t = 1 \text{ V}$ and is $\pm 1 \text{ V}$ away from triode region.



$$I_D = \frac{kn'}{2} \left(\frac{W}{L} \right) (V_{OV})^2$$

$$\therefore 2 = \frac{0.25}{2} \times (V_{OV})^2 \Rightarrow 4 = \frac{0.25}{100} \times (V_{OV})^2$$

$$V_{OV} = \underline{\underline{4V}}$$

$$2 = \frac{1}{100} (V_{OV})^2$$

$$V_{OV} = 4V_{II}$$

$$V_{DS} = V_{OV} + 1 = 5V$$

$$V_D - V_S = 5V$$

$$\therefore V_S = 1V_{II}$$

$$* V_{OV} = V_{GS} - V_{th} = V_G - V_S - V_t$$

$$\therefore V_G = V_{OV} + V_S + V_t = 6V_{II}$$

$$\frac{V_{DD} - V_D}{I_D} = R_D$$

$$\therefore R_D = \frac{4}{2mA} = 2k\Omega$$

similarly

$$R_S = 0.5k\Omega$$

$$* V_G = \frac{V_{DD} \cdot R_2}{R_1 + R_2}$$

$$\frac{6}{10} = \frac{R_2}{R_1 + R_2}$$

$$\cancel{V_{DD}} \quad \oplus \quad V_{DD} = (R_1 + R_2) \times 10 \mu A$$

$$\therefore 10 = R_1 + R_2 \times 10 \mu A$$

$$\therefore R_1 + R_2 = 1M\Omega \quad \text{--- (1)}$$

$$\frac{6}{10} = \frac{R_2}{1M\Omega}$$

$$\therefore R_2 = 0.6M\Omega$$

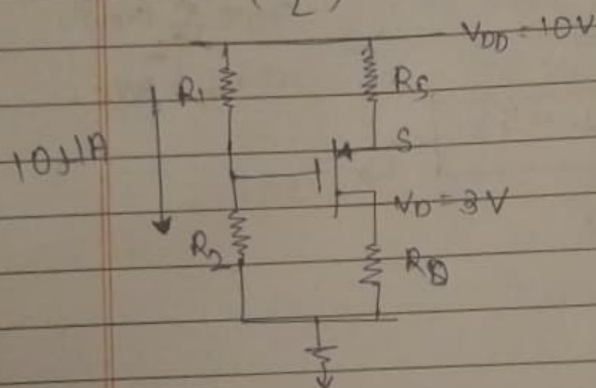
$$\therefore R_1 = 0.4M\Omega$$

Q.

Design a circuit such that transistor is biased ~~800~~ 1 V from edge of triode region

$$I_D = 1 \text{ mA}, \quad V_{DD} = 10 \text{ V}, \quad V_D = 3 \text{ V}, \quad V_t = -1$$

$$K_p' \left(\frac{W}{L} \right) = 0.45 \text{ mA/V}^2$$


Triode Region Condition :-

$$V_{GS} - V_t = V_{DS} \quad V_{DS} = V_{OV} - 1$$

$$I_D = \frac{K_p'}{2} \left(\frac{W}{L} \right) (V_{OV})^2$$

$$1 = \frac{0.45}{2 \times 100} (V_{OV})^2$$

$$\frac{200}{25} = (V_{OV})^2 \quad \rightarrow \text{p.m.s.}$$

$$V_{OV} = -2.82 \text{ V}_{||}$$

$$V_{DS} = -2.82 - 1$$

$$= -3.82 \text{ V}_{||}$$

$$V_D - V_S = -3.82$$

$$-3 + 3.82 = V_S$$

$$V_S = 6.82 \text{ V}_{||}$$

$$V_{GS} - V_{th} = V_{OV}$$

$$V_G - V_S - V_{th} = V_{OV}$$

$$= V_G = -2.82 + 6.82 - 1$$

$$= 3V_{II}$$

$$\frac{V_{DD} - V_S}{I_D} = R_S$$

$$\therefore R_S = \frac{10 - 6.82}{1 \text{ mA}} = 4.18 \text{ k}\Omega$$

$$R_D = 3 \text{ k}\Omega$$

$$V_G = \frac{V_{DD} R_2}{R_1 + R_2}$$

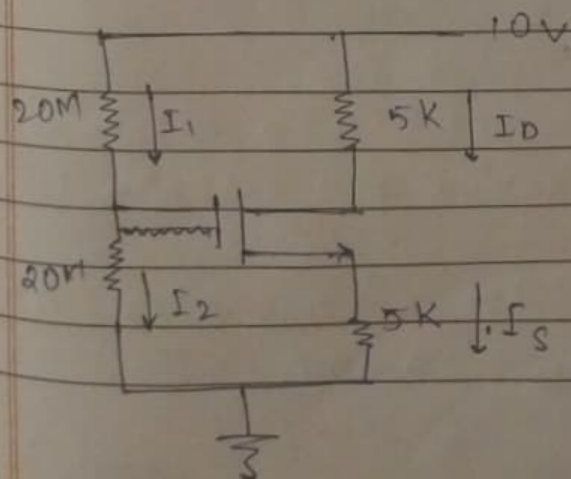
$$\frac{3}{10} = \frac{R_2}{R_1 + R_2}$$

$$V_{DD} = 10 \text{ V} \Rightarrow R_1 + R_2 = 1 \text{ M}\Omega$$

$$R_2 = 0.3 \text{ M}\Omega$$

$$R_1 = 0.7 \text{ M}\Omega$$

Q. In the circuit $V_t = 0.5 \text{ V}$, $K'_n \left(\frac{W}{L} \right) = 0.5 \text{ mA/V}^2$, $V_{DD} = 10 \text{ V}$, find V_G , V_D , V_S , I_1 , I_2 , I_D , I_S .



$$I_D = \frac{0.5 (V_{OV})^2}{2}$$

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$$

$$V_G = \frac{20 \times 10}{40}$$

$$V_G = 5V_{II}$$

$$I_D = \frac{K_n(W/L)}{2} (V_G - V_S - V_t)^2$$

$$V_S - 0 = I_D(5)$$

$$\therefore V_S = 5 I_D$$

$$I_D = \frac{0.5}{2} \left(5 - 5 I_D - \frac{1}{0.5} \right)^2$$

$$2 \times 4 \times I_D \therefore 2 I_D = \frac{1}{2} (4 - 5 I_D)^2$$

$$4 I_D = 16 + 25 I_D^2 - 40 I_D$$

$$25 I_D^2 - 44 I_D + 16 = 0$$

$$I_D = 0.59 \text{ mA} \quad \text{Lower value}$$

$$I_S = 0.59$$

$$I_1 = I_2 = \frac{10}{20+20} = 0.25 \text{ mA}$$

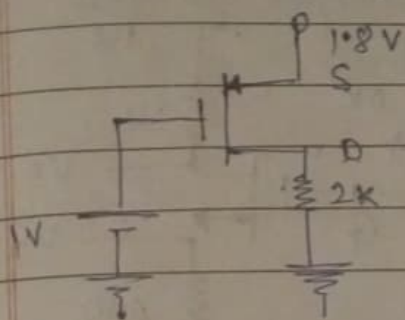
$$10 - V_D = I_D R_D$$

$$10 - I_D R_D = V_D$$

$$V_D = 7.04 \text{ V}$$

$$V_S = I_D \times 5k$$

Q. what ^{value} ~~voltage~~ of $\left(\frac{W}{L}\right)$ places the device in saturation?



$$K_p' = 100 \mu A/V^2$$

$$V_t = -0.4V$$

$$\left(\frac{W}{L}\right) = ?$$

pmos saturatⁿ condⁿ:-

$$V_{DS} < V_{GS} - V_{th}$$

>> At the verge of saturatⁿ

$$V_{DS} = V_{GS} - V_t = 1 - 1.8 + 0.4 = -0.4V_{||}$$

$$V_D - V_S = -0.4$$

$$V_D = 1.4V_{||}$$

$$I_D = \frac{V_D}{2k} = \frac{1.4}{2} = 0.7mA$$

$$I_D = \frac{K_p'}{2} \left(\frac{W}{L}\right) (V_{GS} - V_{th})^2$$

$$0.7 = \frac{100}{2} \times \left(\frac{W}{L}\right) (0.16)$$

$$= \frac{100 \times 0.16}{2} \times \left(\frac{W}{L}\right)$$

$$\frac{1.4 \times 10^{-3}}{0.16 \times 10^{-4}} = \frac{W}{L}$$

$$\frac{W}{L} = \underline{\underline{87.5}}$$

Q. If $\mu_n C_{ox} = 100 \mu A/V^2$, $V_t = 0.5V$, $\frac{W}{L} = \frac{5}{0.18}$

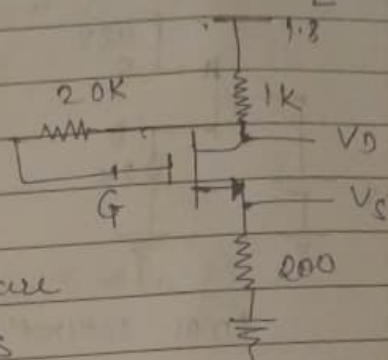
And Drain current

No current can flow from gate

$\therefore V_G$ and V_D are shorted. $\therefore 20K$ is

$$I_g = 0$$

$$\therefore V_G = V_D$$



$$I_D = \frac{K'}{2} \left(\frac{W}{L} \right) (V_{GS} - V_{th})^2$$

$$= \frac{K_D}{2} \left(\frac{W}{L} \right) (V_D - V_S - V_{th})^2$$

$$1.8 - V_D = I_D (1)$$

$$\therefore V_D = 1.8 - I_D \quad \text{--- (1)}$$

$$V_S = 0.2 I_D \quad \text{--- (2)}$$

$$I_D = \frac{0.100}{2} \times \frac{5}{0.18} (1.8 - I_D - 0.2 I_D - 0.5)^2$$

$$= \frac{0.25}{0.36} [1.3 - 1.2 I_D]^2$$

$$= 0.6944 [1.69 + 1.44 I_D^2 - 3.12 I_D]$$

$$= 0.6944 [1.69 + 1.44 I_D^2 - 3.12 I_D]$$

$$= 0.13389 [1.69 + 1.44 I_D^2 - 3.12 I_D]$$

$$I_D = 0.234723 + 0.338 I_D^2 - 1.05456 I_D$$

$$0.338 I_D^2 - 2.05456 I_D + 0.234723 = 0$$

$$I_D = 0.55 mA$$

Quiz Questⁿ:-

- Q. The nmos transistors in circuit are identical in all aspects ~~except~~ except their gate width. Find gate width w_1 & w_2 . If $I_0 = 200 \mu A$,

$$V_{DD} = 5V, V_1 = 2.5V, V_2 = 1V, V_L = 0.5V,$$

$$L_1 = L_2 = L, K_n' = 100 \mu A/V^2$$

$$I_D = \frac{K_n'}{2} \left(\frac{w}{L} \right) (V_{GS} - V_{th})^2$$

$$= \frac{K_n'}{2} \left(\frac{w}{L} \right) (V_1 - V_L - V_t)^2$$

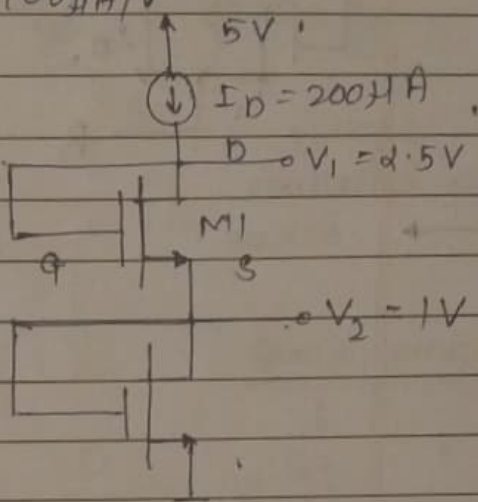
$$200 \mu A = \frac{K_n'}{2} \left(\frac{w_1}{L} \right) (1)^2$$

$$200 \mu A = \frac{100 \mu A/V^2}{2} \left(\frac{w_1}{L} \right)$$

$$\frac{w_1}{L} = 4$$

$$I_D = \frac{K_n'}{2} \left(\frac{w}{L} \right) (1 - 0 - 0.5)^2$$

$$\therefore w = 16L$$



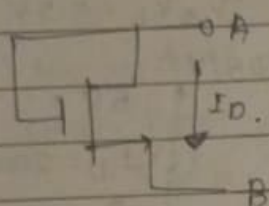
- Q. nmos transistor $C_{ox} = 10 \text{ fF}/\mu\text{m}^2$, $w = 100 \mu\text{m}$, $L = 0.8 \mu\text{m}$, $V_t = 0.5 \text{ V}$, if $V_{GS} = 1.5$ total charge stored in the channel when V_{DS} is equal to 0 is.

$$Q = C_{ox} w L (V_{GS} - V_{th})$$

$$C_{ox} = \frac{10 \times 10^{-15} \text{ F}}{(10^{-6})^2 \text{ m}} = 10 \times 10^{-3} \text{ F/m}^2$$

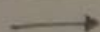
$$Q = 10 \times 10^{-3} \times 100 \times 10^{-6} \times 0.8 \times 10^{-6} \times (1.5 - 1) = 800 \text{ fC}$$

Q. For the circuit shown calculate incremental impedance between a & b assume nmos is operating with $I_D = 1\text{mA}$ $\lambda = 0$ & $K_n' \left(\frac{W}{L} \right) = 0.5\text{mA/V}^2$



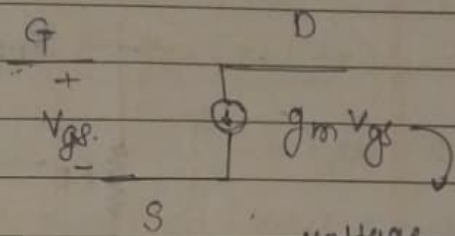
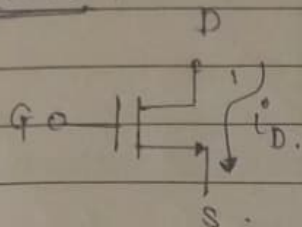
$$K_n' \left(\frac{W}{L} \right) = 0.5\text{mA/V}^2$$

$$I_D = 1\text{mA}$$



Small Signal equivalent :-

(1) NMOS :-

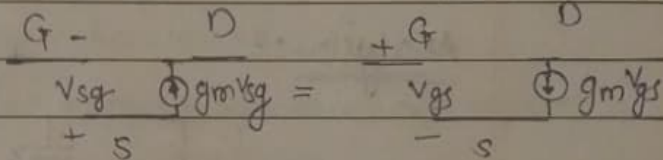
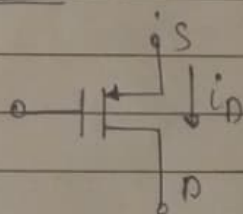


voltage controlled current source.

Source is grounded as nmos is working for logic 0.

Same as nmos.

(2) PMOS.



Configuration :-

drawing small signal model is down

- (1) Common Source Config. i/p & o/p.
- (2) Common Drain Config. ——— Drain ———
- (3) ——— P Gate ——— Gate ———

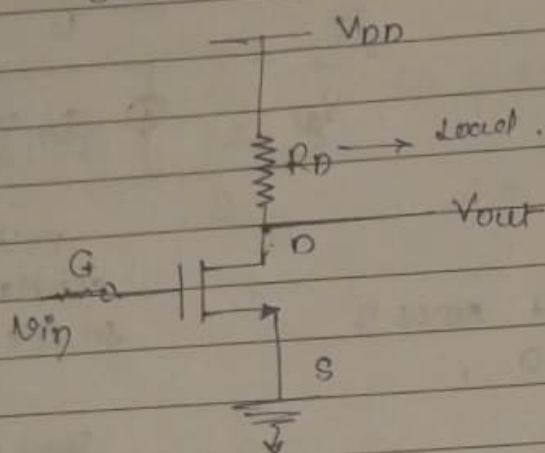
Rule to Draw Signal Model :-

- (1) Short all DC voltage source
- (2) open ——— & current source

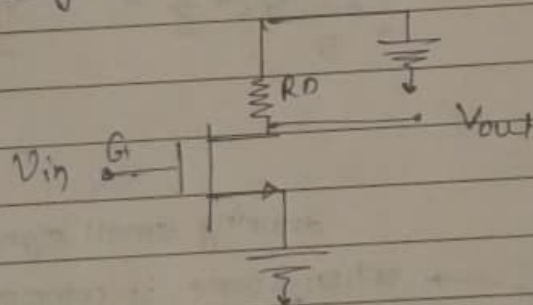
Small signal model tells how circuit will behave when only a AC source is applied.

Common Source Amplifier.

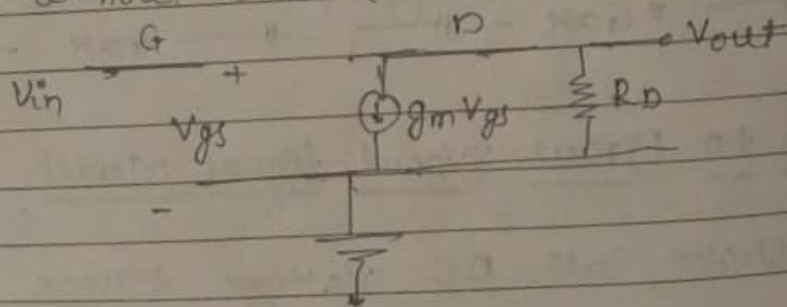
Source terminal is grounded.



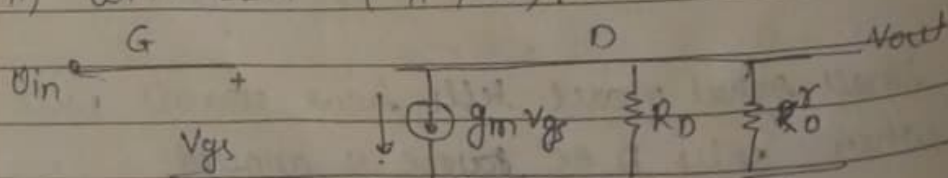
Analysis :-



i) without CLM. ($\lambda = 0$)



ii) With CLM ($\lambda \neq 0$).



$$r_O = \frac{1}{\lambda I_D}$$

i) $Z_{in} = \infty$

ii) $Z_{out} = R_D \parallel r_o$

indicates phase reversal.

iii) $A_v = \frac{V_{out}}{V_{in}} = \frac{-g_m V_{gs} (R_D \parallel r_o)}{V_{gs}} = -g_m (R_D \parallel r_o)$

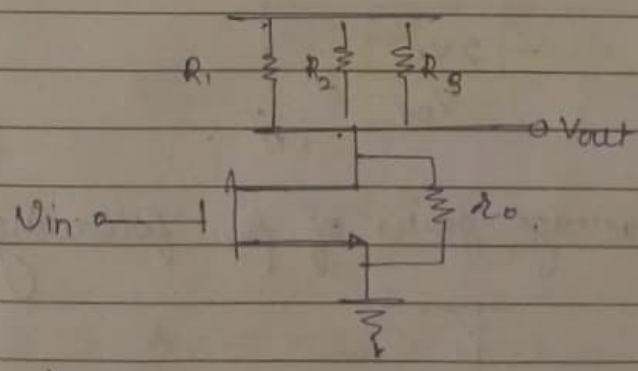
current is flowing from $g_m V_{gs}$ so $V_{out} = -g_m V_{gs} \times (R_D \parallel r_o)$

Conclusions :-

- ① $Z_{out} =$ Resistance ~~same~~ connected to drain.
- ② $Gain = -g_m \times \text{Resistance}$.
 $\hookrightarrow$ transconductance of input transistor.

Q.

Find Z_{in} , Z_{out} , A_v for :- ($\lambda \neq 0$)



$Z_{in} = \infty$

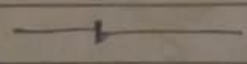
$Z_{out} = R_1 \parallel R_2 \parallel R_3 \parallel r_o$

$A_v = -g_m [R_1 \parallel R_2 \parallel R_3 \parallel r_o]$

Nmos :- standard values

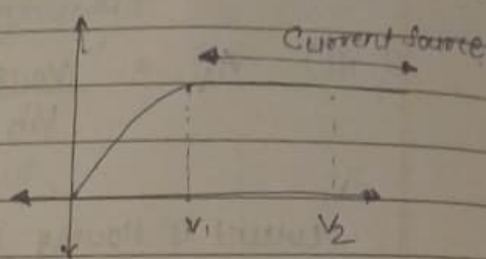
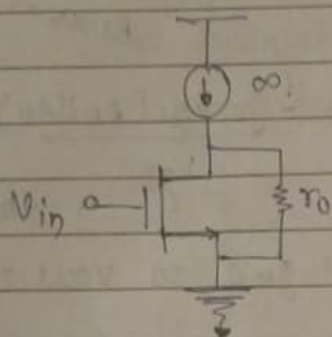
$\lambda_N = 0.1 \text{ V}^{-1}$

Pmos :-



$\lambda_P = 0.2 \text{ V}^{-1}$

Quiz
Intrinsic gain of MOSFET :-



$$R_D = \frac{\Delta V}{\Delta I} = \infty$$

$\infty \rightarrow$ change in current is 0

$$A_V = -g_m [r_o || \infty]$$

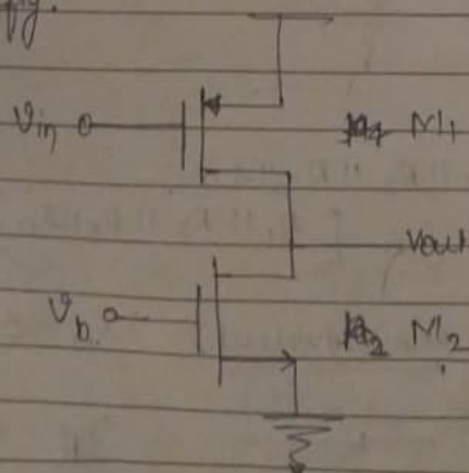
$$= -g_m r_o$$

W.K.T $g_m = \frac{2I_D}{V_{OV}}$, $r_o = \frac{V_A}{I_D}$

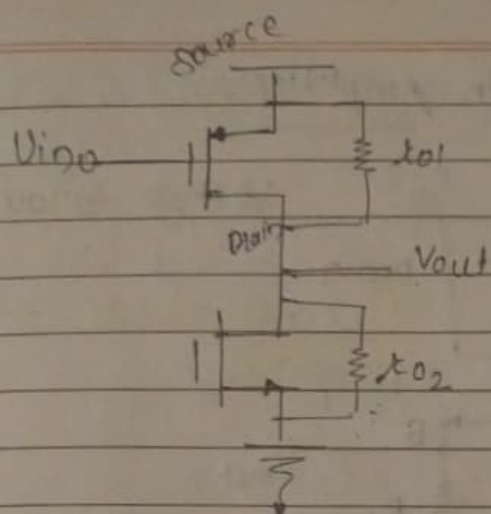
$$A_V = -\frac{2I_D}{V_{OV}} \times \frac{V_A}{I_D}$$

$$A_V = -\frac{2V_A}{V_{OV}}$$

Q2. Find the voltage gain of the following config.



$$A_V = ?$$

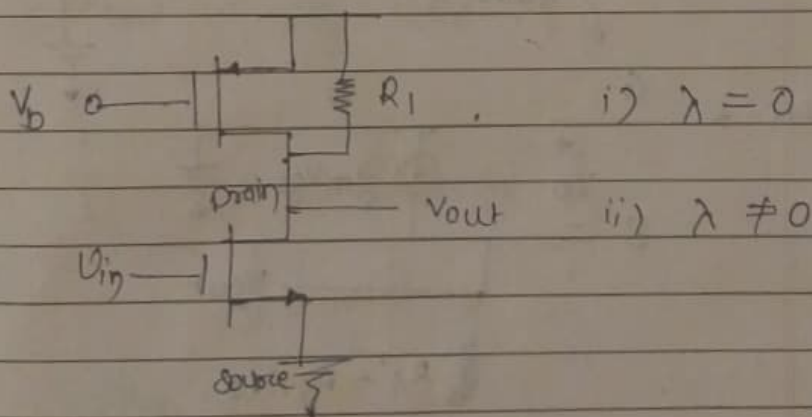


Source is grounded, $\therefore$ CS amplifier.

$$A_v = -g_m (\text{Resistor connected to drain})$$

$$= -g_m (r_{o1} \parallel r_{o2})_{||}$$

②



i) $\lambda = 0$

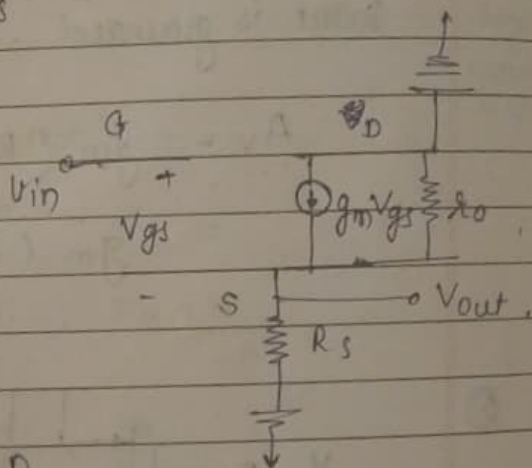
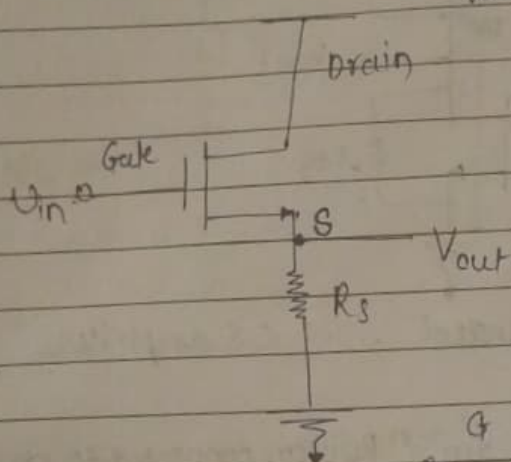
ii) $\lambda \neq 0$

i) $A_v = -g_m R_1$

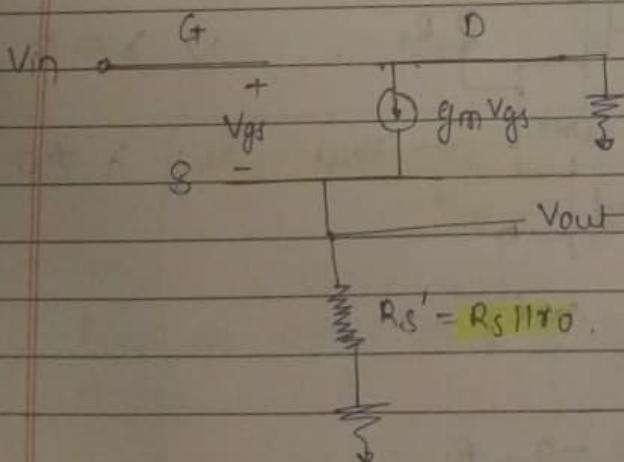
ii) $A_v = -g_m [R_1 \parallel r_{o1} \parallel r_{o2}]$

Common Drain Amplifier

Voltage follower.



same as.



$$Z_i = \infty$$

$$A_v = \frac{V_{out}}{V_{in}} = \frac{g_m V_{gs} R_s'}{V_{gs} (1 + g_m R_s')}$$

$V_{in} \Rightarrow$ Voltage at gate - Voltage at source.

Date _____
Page _____

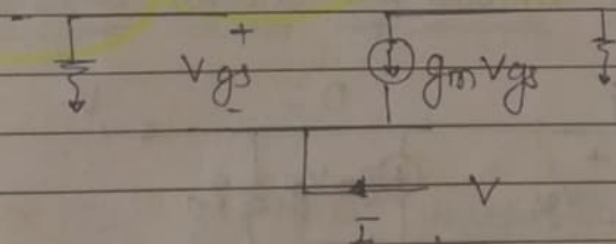
$$A_v = \frac{V_{gs} \times g_m \times R_s'}{V_{gs} (1 + g_m R_s')}$$

$$\Rightarrow A_v = \frac{g_m R_s'}{1 + g_m R_s'} \quad (V_{out} \approx V_{in})$$

since $g_m R_s' \gg 1 \therefore A_v \approx 1$

Z_{out} calculation

- ① Short V_{in} (Ground V_{in})
- ② Remove load
- ③ Apply V to o/p terminal ; find V/I
- ④ $Z_{out} = \frac{V}{I} \parallel R_s$

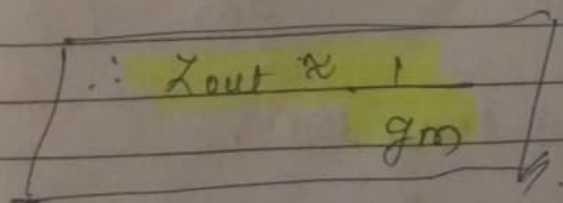


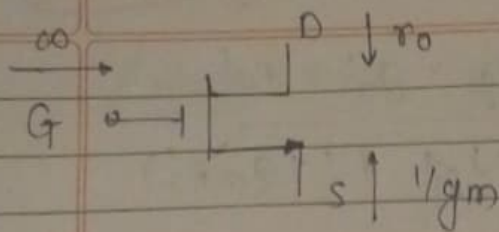
$$V = -V_{gs}$$

$$I = -g_m V_{gs}$$

$$\therefore \frac{V}{I} = \frac{1}{g_m}$$

$$Z_{out} = \frac{1}{g_m} \parallel R_s' = \frac{1}{g_m} \parallel R_s \parallel r_o$$



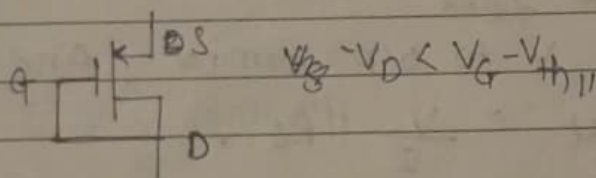
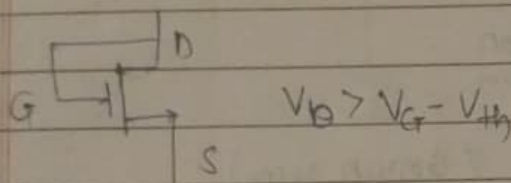


Looking from source,
Resistance is $\frac{1}{g_m}$

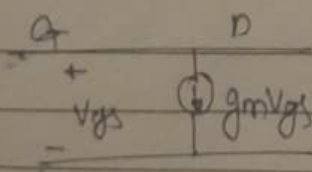
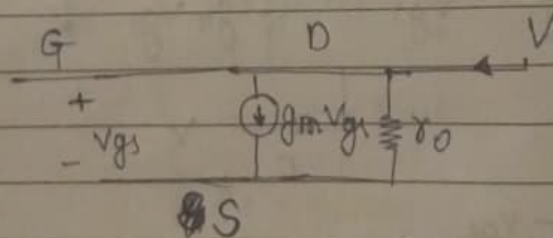
Looking from Drain,
Resistance is r_o

Looking from Gate resistance is ∞ .

Diode Connected Load



IMP Resistance offered by a diode connected load



$$V = v_{gs}$$

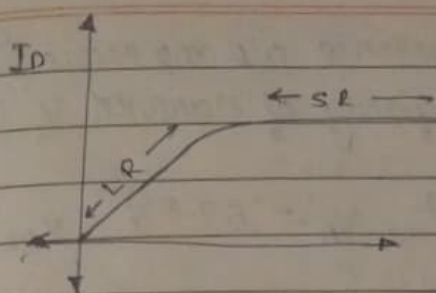
$$I = g_m v_{gs}$$

$$\frac{V_{gs}}{I} = \frac{1}{g_m}$$

$$\therefore Z = \frac{1}{g_m} \parallel r_o \approx \frac{1}{g_m}$$

Impedance is very low.

Quiz:- which of the following ~~works~~ works as current source? Ans:- Config. which offers $Z = \infty$ Trans

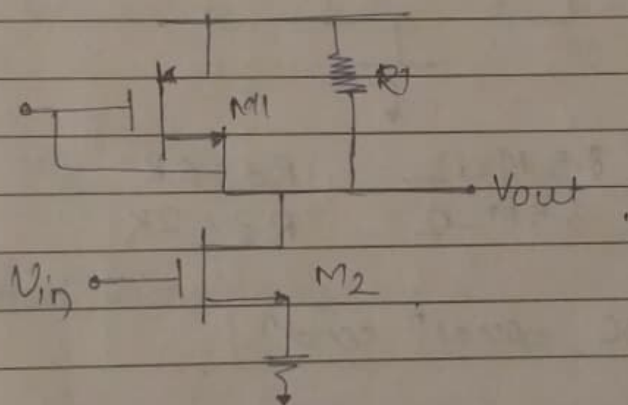


In saturation Region
 $Z \rightarrow \infty$

But in diode connected load in saturation, Z is very less.

Diode connected load will not offer infinite impedance, $\therefore$ It cannot be used as current source.

Q. Quiz:- Find A_V of the following ckt.



$\Rightarrow$ M2 is input transistor,
source is grounded.

$\therefore$ It is common source amplifier.

$$\therefore A_V = -g_{m2} \left[r_{o1} \parallel r_{o2} \parallel R_1 \parallel \frac{1}{g_{m1}} \right]$$

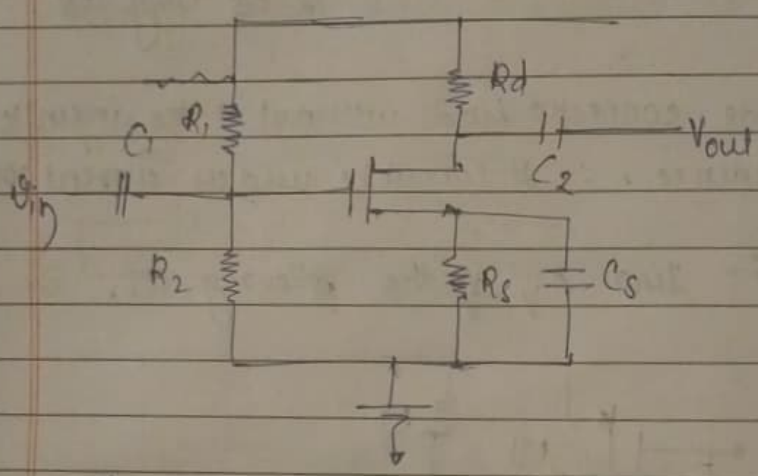
Also M1 is diode connected load, $\therefore$ Its resistance too will be considered.

$$\therefore A_V = -\frac{g_{m2}}{g_{m1}}$$

Q. IMP

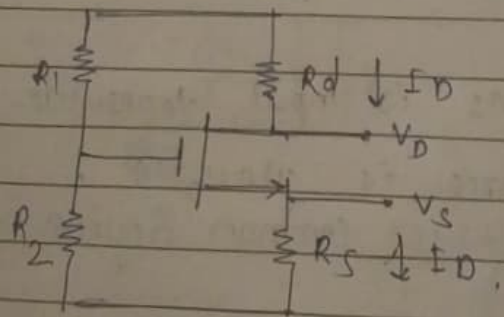
Find gain, input impedance o/p impedance & max signal handling capacity of transistor $V_t = 1V$

$k_m' \left(\frac{W}{L} \right) = 2mA/V^2$, $V_A = 67.5V$, $V_{DD} = 24V$



$R_1 = 8.5M\Omega$ $R_d = 6K$
 $R_2 = 3.5M\Omega$ $R_s = 2K$

① DC operatⁿ condⁿ



$V_G = \frac{V_{DD} R_2}{R_1 + R_2} = 7V$

$R_G = \frac{R_2 R_1}{R_1 + R_2} = 2.47M\Omega$

$$I_D = \frac{kn'(\frac{W}{L})}{2} \left[\frac{V_G}{2} - V_S - V_T \right]^2$$

$$I_D = \frac{kn'(\frac{W}{L})}{2} [7 - 2I_D - 1]^2$$

$$I_D = \frac{2 \text{ mA/V}^2}{2} [7 - 2I_D - 1]^2$$

$$I_D = 7 \cdot (6 - 2I_D)^2$$

$$= 36 + 4I_D^2 - 24I_D$$

$$4I_D^2 - 24I_D + 36 - I_D = 0$$

$$4I_D^2 - 25I_D + 36 = 0$$

$$I_D = 2.25 \text{ mA}, \quad I_D = 4 \text{ mA} \quad X$$

$$* \quad V_D = V_{DD} - I_D R_D = 24 - 2.25 \times 6 = 10.5 \text{ V}$$

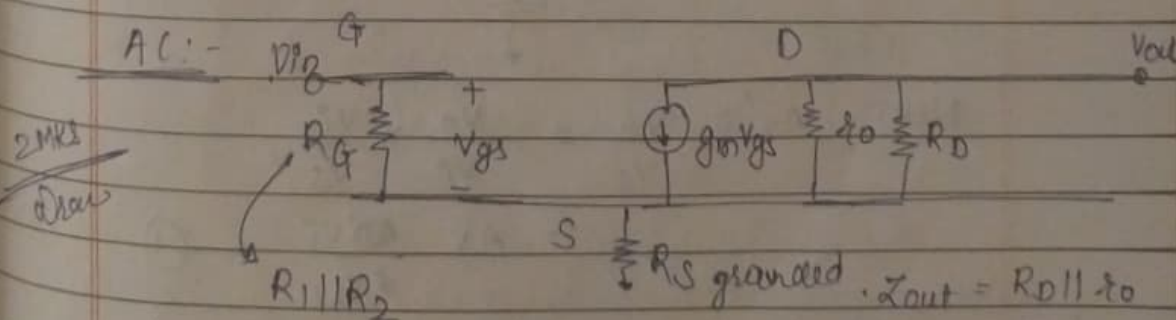
$$V_S = I_D R_S = 4.5 \text{ V}$$

* Small signal Analysis.

$$g_m = \sqrt{2 I_D kn'(\frac{W}{L})} = 3 \text{ mA/V}$$

$$r_o = \frac{V_A}{I_D} = 30 \text{ K} - 24$$

AC:-



$$Z_{in} = R_G = 0.47 \text{ M}\Omega$$

$$A_v = -g_m Z_{out} = -15$$

all capacitors are grounded. $\therefore R_S$ looks like it is grounded.

Max Signal Handling Capacity:-

$$V_D \geq V_G - V_t$$

$$V_{D(DC)} + V_{D(AC)} \geq V_{G(DC)} + V_{G(AC)} - V_t$$

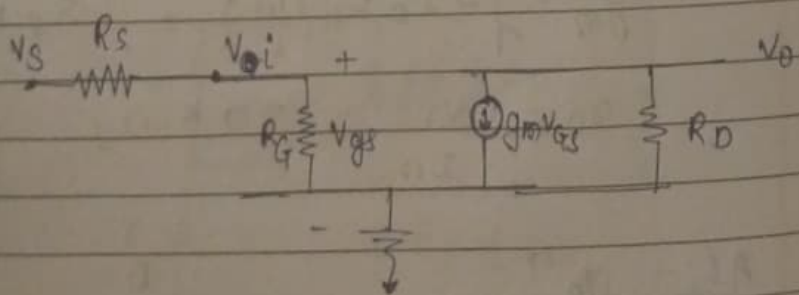
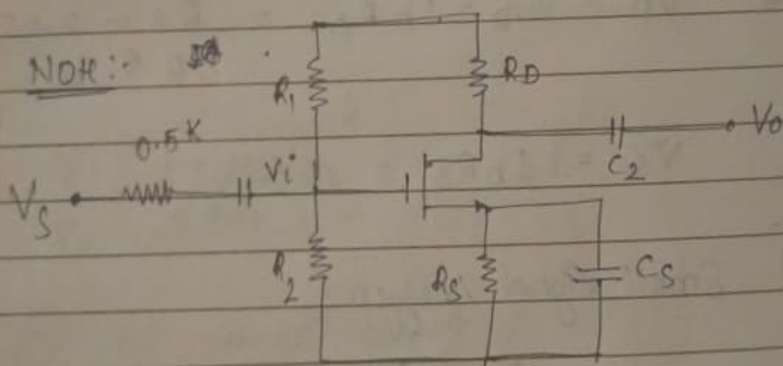
$$10.5 + A_V \hat{V}_i \geq 17 + \hat{V}_i - V_t$$

$$\downarrow$$

$$-15V$$

$$10.5 - 15 \hat{V}_i \geq 17 + \hat{V}_i$$

$$4.5 \hat{V}_i \geq 14 \hat{V}_i$$



$$\text{New } A_V' = \frac{V_O}{V_S} = \frac{V_O}{V_i} \times \frac{V_i}{V_S}$$

$$= A_V \frac{V_i}{V_S} \quad \text{--- (1)}$$

from small signal analysis,

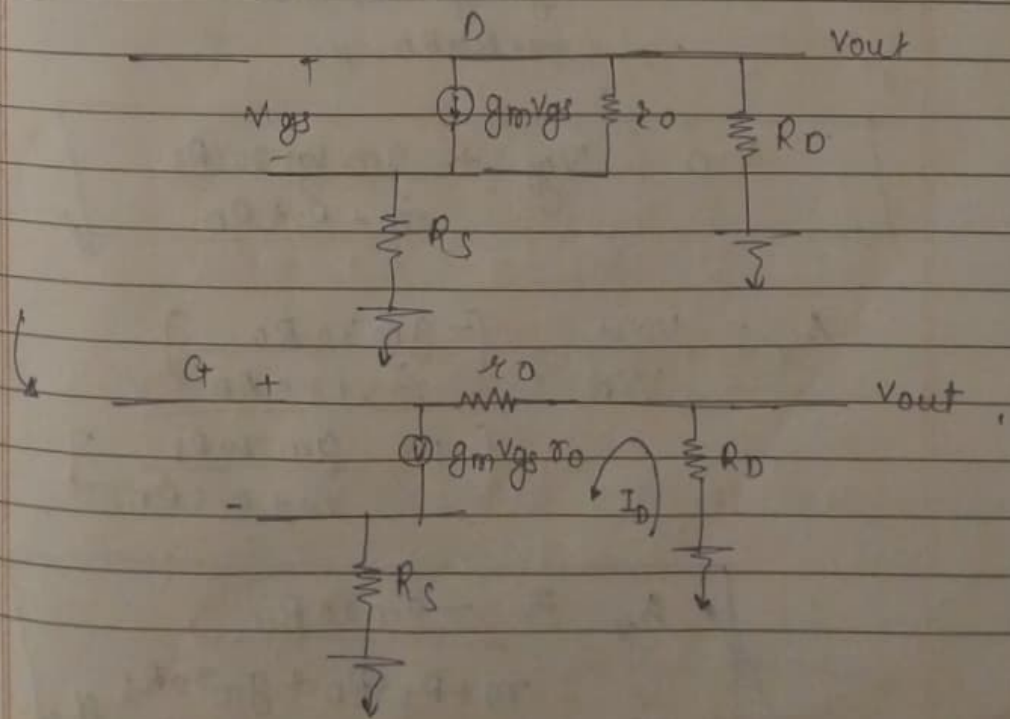
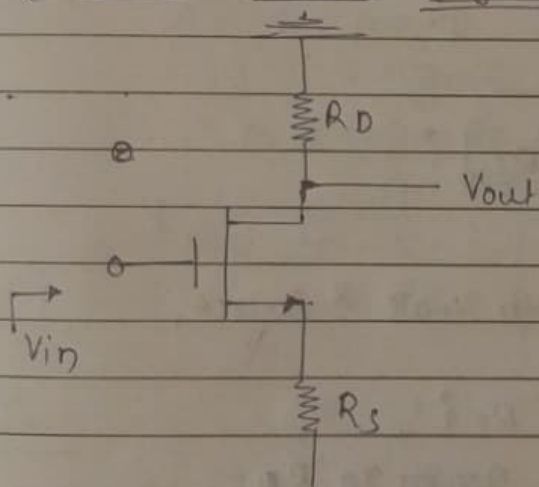
$$V_i = \frac{V_S R_G}{R_S + R_G} \Rightarrow \frac{V_i}{V_S} = \frac{R_G}{R_S + R_G}$$

$$A_v' = A_v \times \frac{R_G}{R_S + R_G}$$

Instead R_D , it can be replaced diode transistor
 $\therefore$ It will have an resistance of $\frac{1}{g_m}$

Similarly R_D can be replaced with current source, $R = \infty$ but $A_v = g_m (r_{o1||\infty}) = g_m r_o$

Common Source Degeneratⁿ



$$i = \frac{g_m v_{gs} r_o}{r_o + R_s + R_D}$$

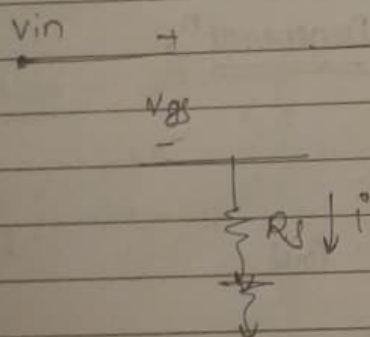
voltage source

All resistance.

$$V_{out} = -i R_D$$

current drop across R_D and R_D is grounded $\therefore -v_o$.

$$\therefore V_{out} = \frac{-g_m v_{gs} r_o R_D}{r_o + R_s + R_D}$$



Applying KVL to Gate & Source,

$$\begin{aligned} V_{in} - V_{gs} &= R_s i \\ &= \frac{g_m v_{gs} r_o R_D}{r_o + R_s + R_D} \end{aligned}$$

$$\therefore V_{in} = V_{gs} + \frac{g_m v_{gs} r_o R_D}{r_o + R_s + R_D}$$

$$A_v = \frac{V_{out}}{V_{in}} = \frac{\left(\frac{-g_m r_o R_D}{r_o + R_s + R_D} \right)}{\left(1 + \frac{g_m r_o R_D}{r_o + R_s + R_D} \right)}$$

$$A_v = \frac{-g_m r_o R_D}{r_o + R_s + R_D + g_m r_o R_D}$$

$$A_V \approx \frac{-g_m r_o R_D}{g_m r_o R_S} \approx \frac{-R_D}{R_S}$$

For common source degen. $A_V \approx \frac{-R_D}{R_S}$

if $\lambda = 0$, $r_o = \infty$.

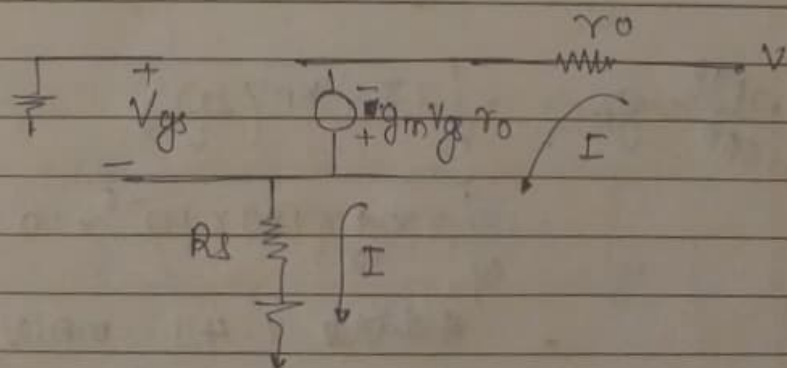
$$\therefore A_V = \frac{-g_m r_o R_D}{r_o + R_S + R_D + g_m r_o R_S} = \frac{-g_m R_D}{1 + \frac{R_S + R_D}{r_o} + g_m R_S}$$

if $\lambda = 0$, $r_o \rightarrow \infty$ $\therefore \frac{R_S + R_D}{r_o} \rightarrow 0$

$$\therefore A_V = \frac{-g_m R_D}{1 + g_m R_S} \approx \frac{-g_m R_D}{g_m R_S}$$

$$\therefore A_V \approx \frac{-R_D}{R_S}$$

Z_{out} :-



$$V + g_m V_{gs} r_o = I (r_o + R_S)$$

considering gate & source, $V_g = 0$ but V_s is having some potential

$$\therefore V_{gs} = -I R_S$$

$\rightarrow V_g$ is grounded $\therefore$ -ve sign.

$$V - \cancel{V_{gs}} g_m r_o I_{AS} = I (r_o + R_s)$$

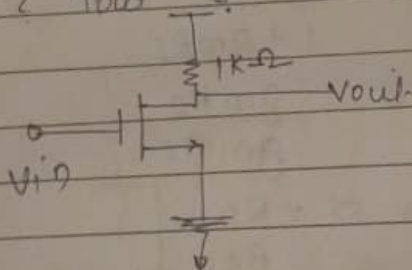
$$\therefore V = I (r_o + R_s + g_m r_o R_s)$$

$$\frac{V}{I} = r_o + R_s + g_m r_o R_s$$

Note:- R_D should be very high value & hence a current source is selected.

Quiz:- $K_n' = 100 \mu A/V^2$ $\frac{W}{L} = 10$, $I_D = 1 \text{ mA}$, $r_o = 10 \text{ K}$

$$A_v = ? \quad R_{out} = ?$$



$$A_v = -g_m [R_D || r_o]$$

$$\frac{1000 \times 10000}{10000}$$

$$g_m = \sqrt{2 I_D K_n' \left(\frac{W}{L} \right)}$$

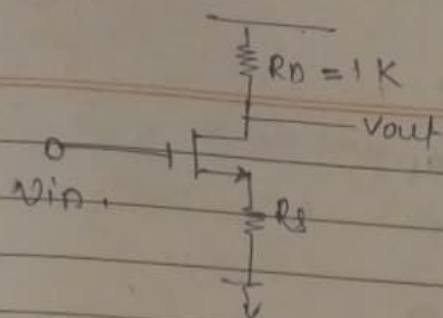
$$= \sqrt{\frac{2 \times 1 \times 100 \times 10^{-6} \times 10}{10^{-3}}}$$

$$= 1.41 \text{ mA/V}$$

$$A_v = -1.41 \times [0.909 \text{ K}]$$

$$= -1.2818$$

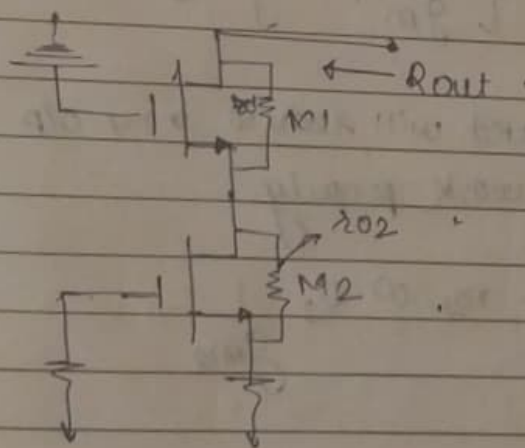
$$= 1.2818$$



$$A_v = \frac{-g_m r_o R_D}{R_D + R_S + r_o + g_m r_o R_S}$$

$$= -1.4 \times 1 \times 10$$

Ques Find the output impedance of the circuit shown below if $r_{o1} = 60K$, $g_{m1} = 3mA/V$, $r_{o2} = 50K$, $g_{m2} = 2mA/V$



In this problem the R_S of source degeneration is replaced with another MOSFET which has resistance of r_{o2} .

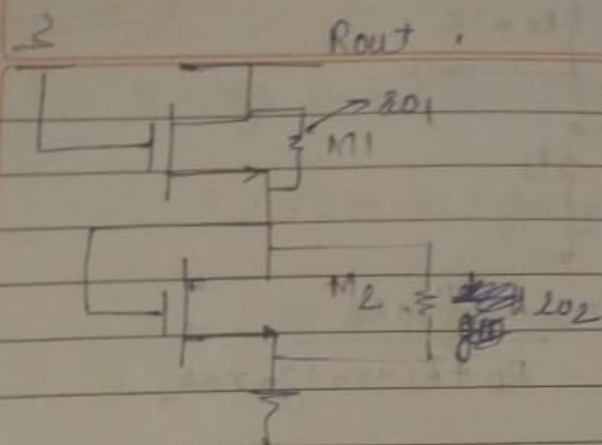
$M1 \rightarrow$ P/p transistor

Remember

$$R_{out} = r_o + R_S + g_m r_o R_S //$$

$$\therefore R_{out} = r_{o1} + R_S + g_{m1} r_{o1} R_S$$

Also $R_S = r_{o2}$ since it is only a resistor.



Here M_2 is diode connected

$\therefore$ Its Resistance is $\left[\frac{1}{g_{m2}} \parallel r_{o2} \right]$

$$\therefore R_{out} = r_{o1} + \left[\frac{1}{g_{m1}} \parallel r_{o1} \right] + g_{m1} r_{o1} \left[\frac{1}{g_{m2}} \parallel r_{o2} \right]$$

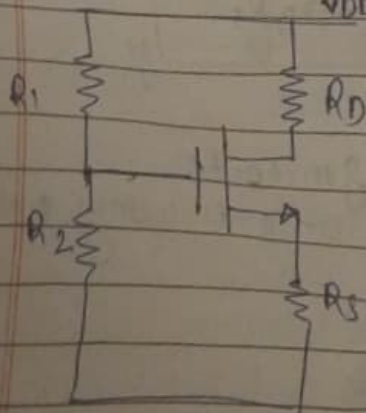
In Diode connected will reduce o/p impedance.

$\therefore$ It won't work properly.

Note:- In (2) if $r_{o2} = \infty$ $R_s = \frac{1}{g_{m2}}$

Q. Design the circuit to obtain $I_D = 1\text{mA}$ assume the transistors are following parameters.

$$V_{DD} = 10\text{V}$$



$$I_D = 1\text{mA}$$

$$K_n' \left(\frac{W}{L} \right) = 2\text{mA/V}^2$$

$$V_t = 1$$

If none of voltage is given.

then $V_D = \frac{2}{3} V_{DD}$

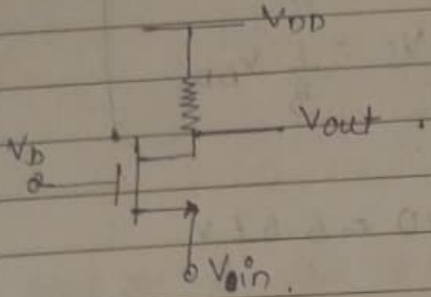
$$V_S = \frac{1}{3} V_{DD}$$

Soln. $V_D = \frac{2}{3} \times 10 = 6.67 \text{ V}$

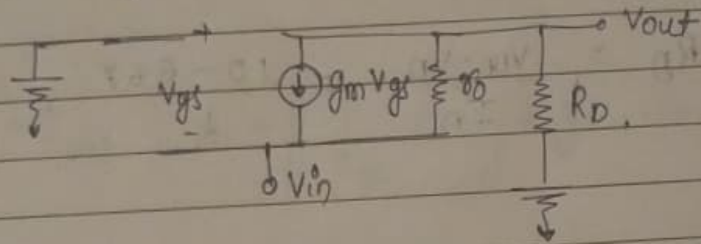
$$V_S = 3.33 \text{ V}$$

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{10 - 6.67}{1} = 3.33 \text{ K}\Omega$$

Common Gate Config :-

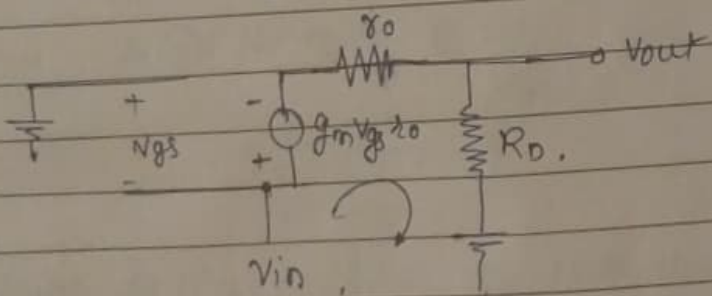


Small signal model :-



$$\text{Voltage gain} = \frac{V_{out}}{V_{in}}$$

$$=$$



Apply KVL to

$$V_{in} - V_{gs} = (R_D + r_o) I_D$$

we know that $V_{gs} = V_g - V_s$
 $= 0 - V_{in}$

$$\therefore V_{in} - g_m (-V_{in}) r_o = (R_D + r_o) I_D$$

$$\therefore V_{in} (1 + g_m r_o) = (R_D + r_o) I_D$$

from Ohm's law,

$$V_{out} = i R_D$$

$$\therefore i = \frac{V_{out}}{R_D}$$

$$\therefore V_{in} (1 + g_m r_o) = i (R_D + r_o)$$

$$V_{in} (1 + g_m r_o) = \frac{V_{out} (R_D + r_o)}{R_D}$$

$$\therefore A_v = \frac{V_{out}}{V_{in}} = \frac{R_D (1 + g_m r_o)}{R_D + r_o}$$

Note:-

if $r_o = \infty$,

$$A_v = \frac{R_D}{r_o} + g_m R_D$$

$$1 + \frac{R_D}{r_o}$$

$$r_o = \infty \left[A_v = g_m R_D \right]$$

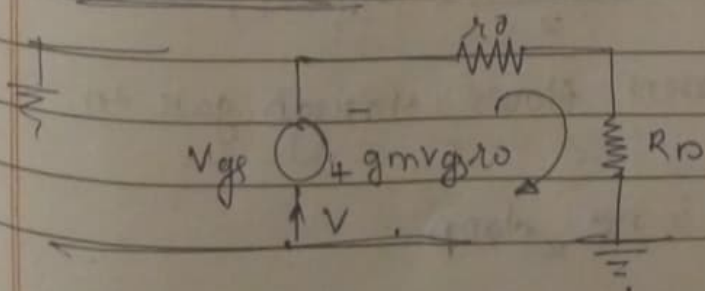
Gain of common gate is same as gain of common source ampli. without phase inversion.

Op impedance :-

(1) Ground all input source.

$$\therefore Z_{out} = r_o \parallel R_D$$

Plp impedance :-



$$V_s - g_m v_{gs} r_o = i (r_o + R_D)$$

$$\therefore \text{Also } v_{gs} = v_g - v_s = -v_s$$

$$\therefore v = [1 + g_m r_o] \cdot i (r_o + R_D)$$

$$\therefore \frac{v}{i} = \frac{r_o + R_D}{1 + g_m r_o} \quad \rightarrow \text{input impedance}$$

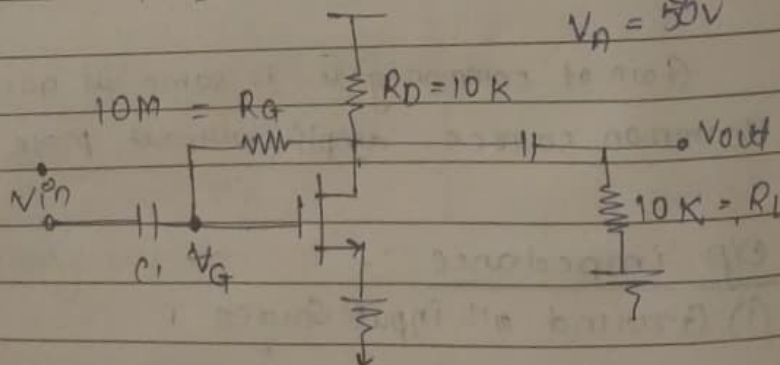
$$\text{if } r_o = \infty$$

$$Z_{in} = \frac{v}{i} = \frac{R_D}{g_m}$$

Q Find small signal voltage gain, Z_{in} , largest allowable signal if $V_t = 1.5V$, $K_n'(W/L) = 0.25 \text{ mA/V}^2$

$$V_{DD} = 15V, R_D = 10K, R_L = 10K, R_g = 10M\Omega$$

$$V_A = 50V$$



$$\rightarrow A_v = ? , Z_{in} = ? , \hat{V}_i = ?$$

$$I_D = \frac{K_n'(W/L)}{2} \times (V_{GS} - V_t)^2$$

NO current flows through gate to drain.

$\therefore$ there is no amp.

$$V_s = 0 \rightarrow \text{grounded}$$

$$\therefore V_G = V_D \quad \text{--- (1)}$$

$$V_{DD} - V_D = I_D (10)$$

$$\therefore V_D = 15 - 10I_D \quad \text{--- (2)} = V_G$$

$$\therefore I_D = \frac{K_n'}{2} \left(\frac{W}{L} \right) (V_{GS} - V_{th})^2$$

$$I_D = \frac{0.25}{2} \times (15 - 10I_D - 1.5)^2$$

$$\therefore 8I_D = (13.5 - 10I_D)^2$$

$$8I_D = 182.25 + 100I_D^2 - 270I_D$$

$$\therefore 92I_D^2 - 270I_D + 182.25 = 0$$

$$\therefore 92I_D^2 - 270I_D + 182.25 = 0$$

$$I_D = 1.05 \text{ mA}$$

$$g_m = \sqrt{2I_D K_n' \left(\frac{W}{L} \right)}$$

$$= 0.72 \text{ mA/V}$$

$$r_o = \frac{V_A}{I_D} = \frac{50}{1.05} = 47.6 \text{ k}\Omega$$

Ans = It is common source amplifier

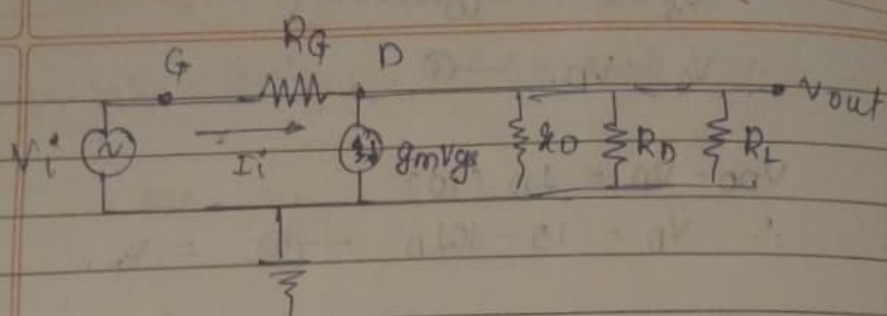
$$A_v = -g_m [R_D || R_L]$$

$$= -g_m [8.26 || R_L]$$

$$= -g_m [4.52]$$

$$= -0.72 \times 4.52$$

$$= -3.25$$



$$Z_{in} = \frac{V_i}{I_i}$$

$$I_i = \frac{V_i - V_{out}}{R_g}$$

$$I_i = \frac{V_i}{R_g} \left[1 - \frac{V_{out}}{V_{in}} \right]$$

$$\therefore \frac{V_i}{R_g} I_i = \frac{V_i}{R_g} [1 - A_v]$$

$$\therefore Z_{in} = \frac{V_i}{I_i} = \frac{R_g}{1 - A_v}$$

$$= 2.33 \text{ M}\Omega$$

* V_i :-

Saturatⁿ: $V_{DS} > V_{DS}$

$$V_{DQ(DC)} + V_D(AC) \geq V_{G(DC)} + V_G(AC) - V_t$$

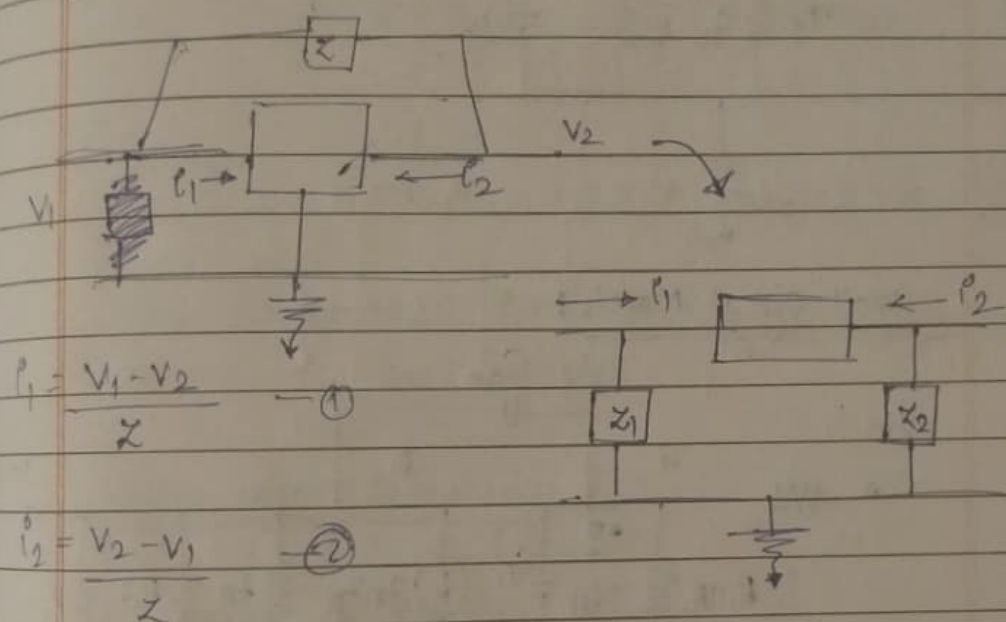
$$(1) \quad 4.5 + 1A_v \hat{V}_i \geq \hat{V}_i + 4.5 - 1.5$$

$$1.5 \geq (1 + 1A_v) \hat{V}_i$$

$$\therefore V_i \leq 0.35$$

Miller's theorem:-

Miller's theorem has used when input impedance b/w i/p terminal & o/p terminal. It is used to split impedance into i/p impedance alone & o/p impedance.



$$i_1 = \frac{V_1 - V_2}{Z}$$

$$i_2 = \frac{V_2 - V_1}{Z}$$

$$i_1 = \frac{V_1}{Z_1} \quad \text{--- (3)}$$

$$i_2 = \frac{V_2}{Z_2} \quad \text{--- (4)}$$

from ① & ③

$$\frac{V_1}{Z_1} = \frac{V_1 - V_2}{Z} \quad \text{--- (5)} \Rightarrow Z_1 = \frac{V_1 Z}{V_1 - V_2}$$

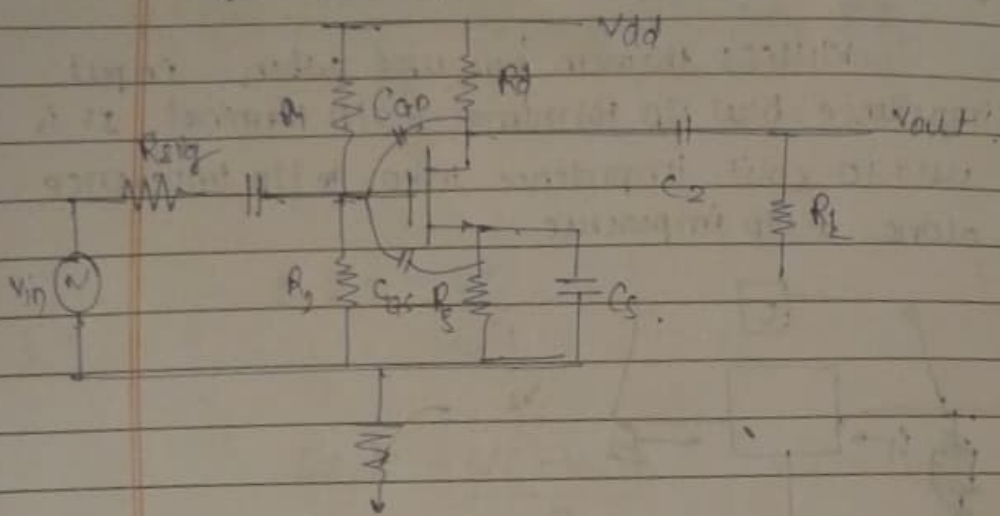
$$\frac{V_2}{Z_2} = \frac{V_2 - V_1}{Z} \quad \text{--- (6)} \Rightarrow Z_2 = \frac{Z}{1 - \frac{V_2}{V_1}}$$

$$\therefore Z_1 = \frac{Z}{1 - A_V}$$

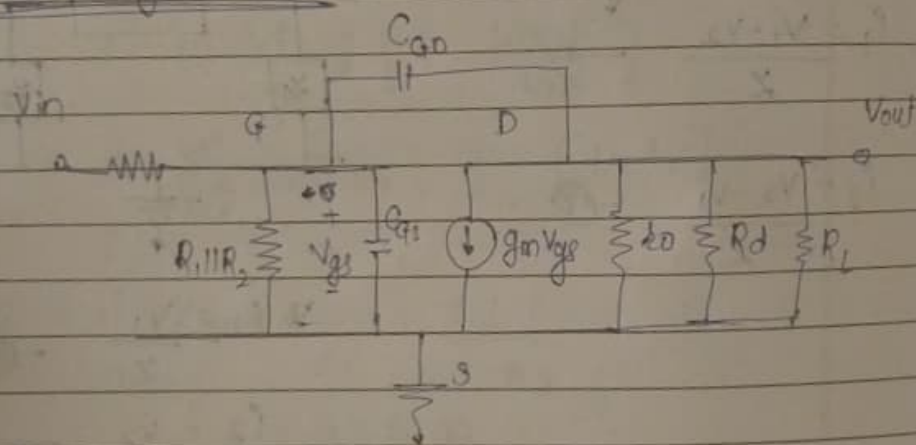
Also

$$Z_2 = \frac{V_2 Z}{V_2 - V_1} = \frac{Z}{1 - \frac{V_1}{V_2}} = \frac{Z}{1 - \frac{1}{A_V}}$$

High Frequency Model of transistor :-



Small signal model :-

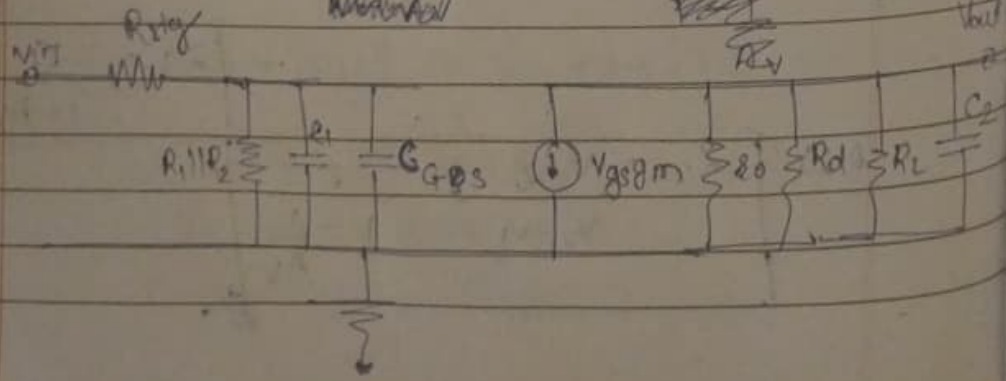


* C_{GD} is connected b/w V_{in} & V_{out} .

Using Miller's theorem,

$$Z_c = \frac{1}{sC}$$

$$C_1 = \frac{C_{GD}(1-A_v)}{A_v} \quad C_2 = \frac{C_{GD}(1-1/A_v)}{1/A_v}$$



1 eqⁿ for CCE & SEC

Lower cutoff Frequency :-

$$(1) f_{LC1} = \frac{1}{2\pi R_{C1} C_1} = \frac{1}{2\pi (R_{sig} + R_1 || R_2) C_1}$$

$$(2) f_{LC2} = \frac{1}{2\pi R_{C2} C_2} = \frac{1}{2\pi R_L (R_d || R_o)}$$

$$(3) f_{cs} = \frac{1}{2\pi R_{C3} C_3} = \frac{1}{2\pi \left(\frac{R_1}{g_m} || R_o || R_3 \right) C_3}$$

for Bandwidth, f_L = Highest of $(f_{LC1}, f_{LC2}, f_{cs})$
usually f_{cs} but calculate.

Higher cutoff Frequency :-

$$C_{out} = C_{ce}$$

$$(1) f_{Hn} = \frac{1}{2\pi R_{eqn} C_{in}}$$

when considering High freq. V_{in} is grounded

$R_{eqn} = R_{sig} || R_1 || R_2$

$$C_{in} = C_{gs} + C_{gd} (1 - A_v) //$$

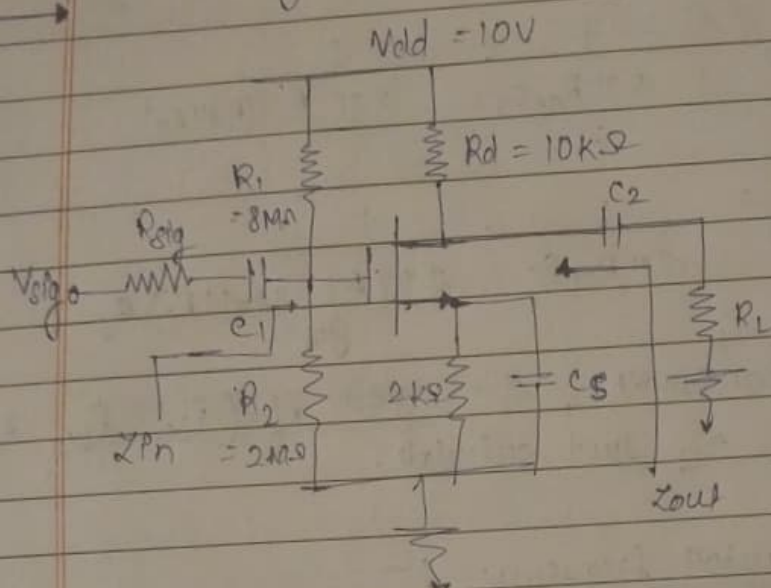
$\therefore R_{sig} || R_1 || R_2$

$$(2) f_{Hout} = \frac{1}{2\pi R_{out} C_{out}}$$

$R_{out} = R_d || R_o || R_L$

f_H = lowest of $(f_{Hn}, f_{Hout}) //$

Q. $K_n'(\mu) = 4 \text{ mA/V}^2$, $V_t = 0.5 \text{ V}$, $V_D = 20 \text{ V}$,
 $C_{gs} = 100 \text{ fF}$, $C_{gd} = 20 \text{ fF}$, find $\frac{V_D}{V_P}$,
 $\frac{V_D}{V_S}$, Z_{in} , Z_{out} , Maximum signal handling
 V_S capacity, f_L , f_H , Bandwidth.



$V_{DD} = 10 \text{ V}$, $R_1 = 8 \text{ M}\Omega$, $R_D = 10 \text{ k}\Omega$, $R_S = 2 \text{ k}\Omega$
 $C_S = 2.5 \mu\text{F}$, $C_1 = 0.01 \mu\text{F}$, $C_2 = 0.1 \mu\text{F}$
 $R_L = 2 \text{ k}\Omega$, $R_{sig} = 30 \text{ k}\Omega$

$$I_D = \frac{K_n'}{2} \left(\frac{W}{L} \right) [V_{GS} - V_{th}]^2$$

$$= \cancel{4 \times 10^{-4}}$$

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$$

$$= \frac{10 \times 2}{10}$$

$$= 2 \text{ V}$$

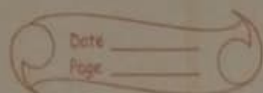
$$I_D = \frac{4^2}{2} [2 - 2I_D - 0.5]^2$$

$$I_D = 2 [1.5 - 2I_D]^2$$

$$= 2 [2.25 + 4I_D^2 - 6I_D]$$

$2 \times 1.5 \times 2$
 $3 \times 1 = 6$

$$\lambda = 0.01 \text{ nmos}$$



4.5

$$I_D = 5 + 8I_D^2 - 12I_D$$

$$8I_D^2 - 13I_D + 5 = 0$$

$$I_D = 0.5$$

$$r_o = \frac{V_A}{I_D} = \frac{20}{0.5} = 40 \text{ k}\Omega$$

$$g_m = \sqrt{2I_D k_n' \left(\frac{W}{L}\right)}$$

$$= \sqrt{2 \times \frac{1}{2} \times 4}$$

$$= 2 \text{ mA/V}$$

$$A_v = \frac{V_o}{V_{in}} = -g_m [R \text{ connected to drain}]$$

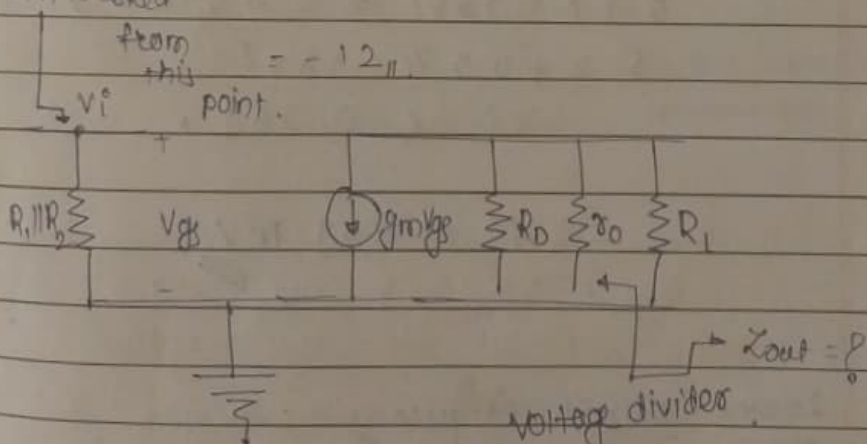
$$= -g_m [R_D || R_L || r_o]$$

$$= -12$$

Z_{in} is asked

from this point.

R_{sig}



voltage divider

$$\frac{V_o}{V_s} = \frac{V_o}{V_i} \times \frac{V_i}{V_s}$$

$$= -12 \times \frac{R_1 || R_2}{R_{sig} + (R_1 || R_2)}$$

$$= -12 \times \frac{1.6 \text{ M}\Omega}{30 + (1.6) \text{ M}\Omega}$$

$$= 11.77$$

Z

In small signal model Z_p is asked from V_i

$$Z_p = R_1 || R_2 = 1.6 \text{ M}\Omega$$

just

Similarly Z_{out} is asked from point next to z_o

Don't consider R_L

$$\therefore Z_{out} = R_d || z_o = 8 \text{ k}\Omega$$

* Max signal handling capacity

$$V_{DS} \geq V_{GS} - V_t$$

$$V_D = 10 - 10I_D = 10 - 10 \times (0.5) = V_D = 5V$$

$$V_{D(DC)} + V_{D(AC)} \geq \frac{2}{3} V_{G(DC)} + \frac{V_{G(AC)}}{3} - V_t$$

$$(10 - 10I_D) + A_v \hat{V}_i \geq 2 + \hat{V}_i - V_t$$

$$5 + (-12) \hat{V}_i \geq 2 + \hat{V}_i - 0.5$$

$$5 - 2 + 0.5 \geq 13 \hat{V}_i$$

$$\therefore \hat{V}_i \leq 0.269 \text{ V}$$

$$\therefore V_i \leq 269 \text{ mV}$$

* Lower cutoff frequency :-

$$f_{Lc1} = \frac{1}{2\pi (R_{Sg} + R_1 || R_2) C_1} = 79.764 \text{ Hz}$$

$$f_{Lc2} = \frac{1}{2\pi R_{Sg} C_1} = \frac{1}{2\pi (R_1 + R_d || z_o) C_2} = 49.78 \text{ Hz}$$

$$R = 24\text{k}\Omega + \frac{10 \times 40}{50}$$

$$= 32 \text{ k}\Omega$$

$$f_{LC2} = \frac{1}{2\pi R_{CS} C_S} = \frac{1}{2\pi \left(\frac{1}{g_m} \parallel R_0 \parallel R_S \right) C_S}$$

$$R = 0.9960 \text{ K}\Omega = 160.76 \text{ Hz}$$

$$\therefore f_L = 160.76 \text{ Hz}$$

Higher cutoff ~~freq~~ frequency.

$$f_{fn} = \frac{1}{2\pi (R_{sig} \parallel R_1 \parallel R_2) C_{in}} = 15.04 \text{ MHz}$$

$$= \frac{1}{2\pi \cdot 29.44 \cdot C_{in}}$$

$$C_{in} = C_{gs} + C_{gd} (1 - A_v) = 100 \text{ pF} + 20 (1 + 12)$$

$$= 100 + 20 \times 13 \text{ pF}$$

$$= 360 \text{ pF}$$

$$f_{Cout} = \frac{1}{2\pi R_{Cout} C_{out}} = 12.24 \text{ MHz}$$

$$R_{Cout} = R_D \parallel R_0 \parallel R_L = 6 \text{ K}\Omega$$

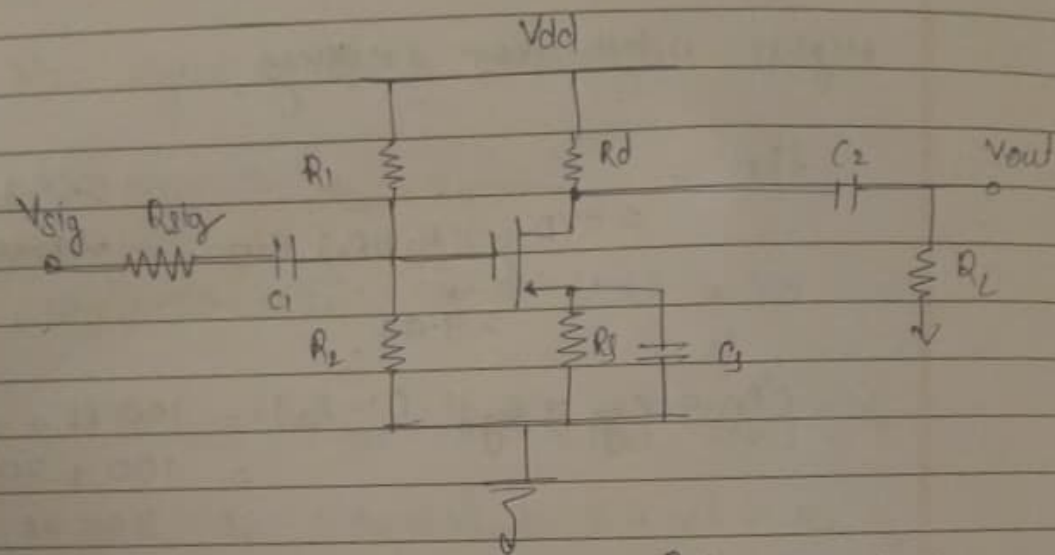
$$C_{out} = C_{gd} \left(1 - \frac{1}{A_v} \right) = 21.67 \text{ pF}$$

$$\text{Bandwidth} = f_H - f_L$$

$$= 15.04 \text{ MHz} - 160.76 \text{ KHz}$$

Q In the CS amplif. circuit $c_{gs} = 1 \text{ pF}$, $c_{gd} = 0.2 \text{ pF}$,
 $\mu_n k_n' (W/L) = 1 \text{ mA/V}^2$, $\lambda = 0$, ($r_o \approx \infty$),
 $I_D = 1 \text{ mA}$, $V_t = 1 \text{ V}$, determine mid frequency
 gains, lower cutoff freq. due to coupling
 & bypass caps., Upper cutoff frequency
 due to Bandwidth.

$\frac{V_o}{V_i} = \frac{V_o}{V_i}$



$R_1 = 5 \text{ M}\Omega$, $R_2 = 5 \text{ M}\Omega$, $R_{sig} = 100 \text{ k}\Omega$,
 $C_1 = C_2 = 1 \mu\text{F}$, $C_3 = 10 \mu\text{F}$, $R_d = 10 \text{ k}\Omega$,
 $R_s = 3 \text{ k}\Omega$, $R_L = 20 \text{ k}\Omega$.

Know -

$$I_D = \frac{k_n' (W/L)}{2} (V_G - V_S - V_{th})^2$$

$$I_D = \frac{1}{2} \left(\frac{V_{DD} R_2}{R_1 + R_2} - 3 I_D - V_t \right)^2$$

$$1 = \frac{1}{2} \left(\frac{V_{DD} \times 5}{10} - 3 \times (1) - 1 \right)^2$$

$$2 = \left(\frac{V_{DD}}{2} - 3 - 1 \right)^2$$

$$2 = \left(\frac{V_{DD}}{2} - 4 \right)^2$$

$$V_{DD} =$$

$$g_m = \sqrt{2 I_D k_n' \left(\frac{W}{L} \right)}$$

$$= \sqrt{2 \times (1) \times 1}$$

$$= 1.41$$

$$A_v = \frac{V_o}{V_i} = -g_m [R_{\text{connected to Drain}}]$$

$$= -1.41 [R_D || R_L]$$

$$=$$